Ex-ante Heterogeneity in Directed Search Games

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Abstract

We study a directed search game in which finitely many workers each apply to one of two firms paying different wages, and each firm hires its most skilled applicant. The standard analysis assumes that all workers draw their skills from one common distribution and focuses on the symmetric equilibrium. We show that this equilibrium is not continuous in ex-ante heterogeneity. Arbitrarily small differences between workers' skill distributions can produce a unique equilibrium that is sharply asymmetric. We obtain three results. First, an equilibrium of the game with identical workers is robust to small ex-ante heterogeneity if and only if it is in pure strategies. This equilibrium exists, is unique up to relabelling the workers, and sorts workers into groups that apply to the high-wage firm above group-specific skill thresholds. Second, when nearby skill distributions are ranked by first-order stochastic dominance, the ranking orders the workers' thresholds, so a worker with a dominated distribution applies to the high-wage firm only at high skill. Third, as the market grows, aggregate application behaviour and interim payoffs in this equilibrium converge to those of the symmetric equilibrium.

1 Introduction

Models of directed search usually assume that workers draw their skills from a common distribution. The assumption is made for convenience. It rests on an implicit belief that workers who are nearly identical ex ante behave nearly as they would if they were exactly identical. We show that in a finite directed search game this belief fails.

Consider two workers, each of whom applies to one of two firms. Firm 1 pays a higher wage than firm 2, and a firm with more than one applicant hires the most skilled. When both workers draw their skills from the same distribution, the familiar symmetric equilibrium of Peters (2010) [3] has each worker apply to firm 1 for sure when his skill is high and randomise between the firms when it is low.

Now let worker 1's skill distribution be slightly better than worker 2's. We find that the game then has a unique, asymmetric equilibrium. Worker 1 applies to firm 1 at every skill level, while worker 2 applies to firm 1 only when his skill is high and to firm 2 otherwise. Section 4.2 shows why. As the difference between the two distributions shrinks to zero, this equilibrium converges to an asymmetric equilibrium of the game with identical workers, not to the symmetric one. The

symmetric equilibrium is therefore not continuous in ex-ante heterogeneity.

We ask which equilibria of the game with identical workers survive small ex-ante heterogeneity. We call an equilibrium robust to small ex-ante heterogeneity, or robust for short, if every game whose skill distributions are close enough to the common one has an equilibrium close to it, up to relabelling the workers. We study this question in the model of Peters (2010) with N workers and two firms.

Our main result, Theorem 1, characterises the robust equilibria. An equilibrium is robust if and only if it is in pure strategies. The pure-strategy equilibrium exists for every number of workers and is unique up to relabelling. In it, workers sort into groups, and each group applies to the high-wage firm exactly when its members' skill exceeds a group-specific threshold. In particular, the symmetric equilibrium is not robust.

The second result identifies which worker takes which role. When the skill distributions are close and ranked by first-order stochastic dominance, the ranking orders the workers' thresholds in every equilibrium (Lemma 1). A worker whose distribution is dominated applies to the high-wage firm only when his skill is high. A worker whose distribution dominates applies there even when his skill is low. Small differences that all market participants recognise, such as the prestige of a worker's school or perceptions attached to a demographic group, therefore select which workers apply aggressively and which apply cautiously.

The third result reconciles the robust equilibrium with the symmetric one in large markets. We let the numbers of high-wage and low-wage firms grow in proportion to the number of workers. Individual strategies in the two equilibria remain very different, but the share of workers at each skill level who apply to the high-wage firm converges to the same limit in both, and so do workers' interim payoffs. For aggregate application behaviour in large markets, the two equilibria therefore make the same predictions.

2 Related Literature

Existing literature on directed search can be separated into two categories (see Wright et al. (2021) [2] for a full review). The first class, usually used in labour and macroeconomics, assumes a continuum of workers and firms and an exogenous matching function. The probability that a given worker obtains a job is determined by the matching function evaluated at the market tightness (worker to firm ratio) of that type of job. Equilibrium is characterised by market clearing.

The second class of model deals with a finite number of workers and firms and explicitly models the strategic interactions amongst the players. In these models, firms post wages and numbers of openings, and the workers then simultaneously choose which firms to apply to. It is usually assumed that the workers can apply to only one job. In the event that more than one worker applies for the same firm, the firm in question uses some predetermined mechanism to decide who gets the job. This framework uses Nash equilibrium as its equilibrium concept.

It has been shown that the second class of models converges to the first class as the number of workers and firms goes to infinity [6]. However, the two frameworks are sufficiently different mathematically that it makes sense to treat them separately.

Player heterogeneity has been studied extensively in the literature. Peters (2010) [3] provided a characterisation of the symmetric equilibrium with finite players when workers are ex-post heterogeneous. Eeckhout and Kircher (2010) [5], using a continuum of players, derived necessary and sufficient conditions for positive assortative matching between workers and sellers (i.e. high skilled workers are matched with high quality firms). This paper studies the same strategic framework as Peters (2010), but with a special focus on the robustness of equilibria.

A common feature across a number of directed search frameworks is that the equilibrium is difficult to solve analytically but easy to construct algorithmically. While there are several variations of the equilibrium construction algorithm, they all exploit the downward recursive properties of the payoff functions. The idea of this algorithm can be traced back to Chade (2006) [7]. Focusing on symmetric equilibria, Camera and Kim (2013, 2014) [4][8] used one such algorithm to establish the uniqueness of the symmetric Nash equilibrium in the presence of ex-post worker heterogeneity. Similarly, Guerrieri et al. (2009) [9] developed an equilibrium construction algorithm for a continuum of players. In this paper, we provide an algorithmic characterisation of the robust equilibrium that is similar in spirit to the ones currently found in the literature, and exploit the downward recursive nature of the payoffs to establish its theoretical properties.

Moving beyond the directed search literature, this paper can also be viewed through the lens of equilibrium refinement[1]. We propose an equilibrium characterisation that satisfies strict perfection and bears certain similarity to essential equilibria[1]. Such robustness properties are particularly relevant for empirical works. Parametric economic models built around "robust" equilibria are more resilient to minor misspecification concerning game structure. In the context of this paper, the misspecification arises from small ex-ante heterogeneity that is known to market participants but unobserved to the analyst.

3 Model

In this section, we introduce the theoretical framework of this paper. We consider a market with N workers and 2 employers. We first consider the symmetric equilibrium under ex-ante identical workers. We then motivate the study of alternative equilibrium characterisations by examining equilibrium behaviours under minute ex-ante heterogeneity.

We index the worker by i and denote worker i's skill by θ_i. θ_i is independently distributed according to some distribution with CDF $F_i(\cdot)$. Throughout this paper, we assume that $F_i(\cdot)$ is twice continuously differentiable everywhere and has support on the interval $[a,b]$ $\forall i \in \{1,...,N\}$. We

[1]Intuitively, an equilibrium is strictly perfect if it is "robust" to perturbations in player strategies. An essential equilibrium is an equilibrium that is "robust" to perturbations in game parameters

denote the vector of skill distributions by $\mathbb{F} = \begin{bmatrix} F_1(\cdot) \\ ... \\ F_N(\cdot) \end{bmatrix} \in [a,b]^N$. We say that workers are ex-ante homogeneous or identical if $F_i(\cdot) = F_j(\cdot) \forall i, j \leq N$, and ex-ante heterogeneous if not. Finally, as is standard in the literature, we use the $-i$ subscript to denote the vector of attributes of all workers other than i.

Worker i observes his own skill realisation θ_i, the offered wages w_1 and w_2, the total number of workers N, and the vector of skill distributions $\mathbb{F}$. He then chooses to apply to one of two firms. Worker i does not observe the skill realisations of other workers nor their application decisions. We denote the realised vector of worker skills by $\theta = \begin{bmatrix} \theta_1 \\ ... \\ \theta_N \end{bmatrix} \in [a,b]^N$.

Firm 1 offers wage w_1 and firm 2 offers wage w_2. We assume W.L.O.G. that $w_1 \geq w_2$. Firms observe the workers' skill realisations perfectly. If the firm only receives a single application, then it hires the worker regardless of the worker's skill. However, if the firm receives more than one application, it hires the worker with the highest realised skill (ties are broken randomly). If the worker is hired by firm 1(2), he receives payoff $w_{1(2)}$. An unemployed worker's payoff is normalised to 0.

We denote worker i's action by $a_i \in \{1,2\}$, where $a_i = 1$ if worker i applies to firm 1, and $a_i = 2$ if worker i applies to firm 2. A strategy is a function mapping from the support of the skill distribution to the probability of applying to a given firm. Formally, we denote worker i's strategy by $\pi_i(\cdot) : [a,b] \to [0,1]$, where

$$\pi_i(x) = \text{Prob}(a_i = 1 | \theta_i = x)$$

A strategy profile is a vector valued function represented as a vector of strategies of length N. The i-th entry corresponds to the strategy of worker i. We denote the strategy profile with capital letter $\Pi = \begin{bmatrix} \pi_1(\cdot) \\ ... \\ \pi_N(\cdot) \end{bmatrix}$, and use Π_{-i} to denote the strategy profile containing the strategies of all workers other than worker i. With a slight abuse of notation, we also define the vector valued function $\Pi(\theta) : [a,b]^N \to [0,1]^N$ as:

$$\Pi(\theta) = \begin{bmatrix} \pi_1(\theta_1) \\ ... \\ \pi_N(\theta_N) \end{bmatrix}$$

Worker i chooses $\pi_i(\cdot)$ to maximise his expected utility given his skill realisation. His payoff from applying to a given firm is simply the wage offered by that firm multiplied by the probability that he gets the job given his skill and the application strategies of other workers. We define agent i's payoff from choosing action a_i given his realised skill level θ_i and the strategies of every other

agent Π_{-i} as $u_i(a_i|\theta_i,\Pi_{-i})$, where:

$$u_i(a_i|\theta_i,\Pi_{-i}) = \begin{cases} w_1 \prod_{h=1,h\neq i}^{N} \left[F_h(\theta_i) + \int_{\theta_i}^{b} (1-\pi_h(\mu))\, dF_h(\mu)\right] & \text{if } a_i = 1 \\ w_2 \prod_{h=1,h\neq i}^{N} \left[F_h(\theta_i) + \int_{\theta_i}^{b} \pi_h(\mu) dF_h(\mu)\right] & \text{if } a_i = 2 \end{cases}$$

And the worker's problem given θ_i and Π_{-i} is defined as:

$$\max_{p\in[0,1]} \{p u_i(1|\theta_i,\Pi_{-i}) + (1-p) u_i(2|\theta_i,\Pi_{-i})\}$$

With $\pi_i(\theta_i)$ being the solution to the above problem.

Notice that for any given θ_i, the value of $\pi_j(\theta_j)$ for $\theta_j \in [a,\theta_i]$ has no impact on worker i's payoff and hence no impact on worker i's problem. This is because a worker never needs to worry about the application decisions of those with skills lower than his. If they apply to the same firm as he does, he can outcompete them; if they apply to a different firm from him, then they do not impact his chances of getting hired. As a result, equilibrium can be easily calculated recursively starting from the equilibrium application decision of workers with the highest skill realisations. Mathematically, if we know the equilibrium values of $\pi_i(\theta_i)$ $\forall i$ and $\forall \theta_i > x$, where $x \in (a,b]$ is some constant, then we can calculate $u_i(1|x,\Pi_{-i})$ and $u_i(2|x,\Pi_{-i})$, and pin down the equilibrium value of $\pi_i(x)$ for all i.

Finally, we denote a game with N workers, two firms offering w_1 and w_2, and skill distributions $\mathbb{F}$ by:

$$\Gamma(N,w_1,w_2,\mathbb{F}) \text{ or } \Gamma(\mathbb{F})$$

when N,w_1,w_2 are unimportant. In particular, we denote the special case in which $F_i(\cdot) = F(\cdot)$ $\forall i$ by:

$$\Gamma(N,w_1,w_2,F(\cdot)) \text{ or } \Gamma(F(\cdot))$$

when N,w_1,w_2 are unimportant. We refer to games where all workers draw their skills from the same distribution as symmetric games and games where workers have different skill distributions as asymmetric games.

4 Equilibria

In the following section, we focus on equilibrium behaviours of the game $\Gamma(N,w_1,w_2,F(\cdot))$. When all workers have the same skill distribution, the game reduces to a simplified case of the one studied in Peters (2010) [3]. We first present the equilibrium characterised by Peters (2010) as a baseline, and then discuss what happens when workers' skill distributions are not exactly identical.

Throughout this paper, we define an equilibrium as:

Definition 1: A strategy profile Π *is an equilibrium of* $\Gamma(N, w_1, w_2, \mathbb{F})$ *if and only if* $\forall i \in \{1, ..., N\}$ *and* $\forall \theta_i \in [a, b]$*:*

$$\begin{aligned} E[u_i(1|\theta_i, \Pi_{-i})|\Pi, \theta_i] &= u_i(1|\theta_i, \Pi_{-i})\pi_i(\theta_i) + u_i(2|\theta_i, \Pi_{-i})(1 - \pi_i(\theta_i)) \\ &\geq \max_{p \in [0,1]} \{u_i(1|\theta_i, \Pi_{-i})p + u_i(2|\theta_i, \Pi_{-i})(1-p)\} \end{aligned}$$

4.1 Symmetric Equilibrium

As shown in Peters (2010), the game always has a unique symmetric Nash equilibrium:

Proposition 1: For any continuously differentiable distribution $F(\cdot)$ *with support on [a,b], the game has a unique symmetric Nash equilibrium* $\Pi^S(\theta) = \begin{bmatrix} \pi_1^S(\theta_1) \\ ... \\ \pi_N^S(\theta_N) \end{bmatrix}$ *where* $\forall i \in \{1, ..., N\}$*:*

$$\pi_i^S(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq F^{-1}\left((\frac{w_2}{w_1})^{\frac{1}{N-1}}\right) \\ \frac{1}{(\frac{w_1}{w_2})^{\frac{1}{N-1}} + 1} & \text{if else} \end{cases}$$

This result follows directly from the work of Peters (2010). Intuitively, if worker i is sufficiently skilled (θ_i is sufficiently large), then it is always optimal for him to apply to firm 1. For workers with low skills, they still apply to firm 1 with positive probabilities because it is always possible that no one else applies to firm 1 and they receive the job.

This equilibrium implies that all workers apply to firm 1 with a constant probability when their skills are below some threshold. While this characterisation is elegant and tractable, it is also extremely unstable. For the sake of simplicity, consider the case when $N = 2$, and suppose worker 1 with $\theta_1 = \theta^* < F^{-1}\left((\frac{w_2}{w_1})\right)$ applies to firm 1 with a probability p that is slightly higher than $\pi_1^S(\theta^*)$. This deviation is incentive compatible since worker 1 is indifferent between any $\pi_1(\theta^*) \in [0, 1]$ by construction. However, worker 2 with $\theta_2 < \theta^*$ would then find it strictly optimal to apply to firm 2 with probability 1. This in turn makes it strictly optimal for worker 1 to apply to firm 1 whenever his skill is below θ^*, resulting in another Nash equilibrium $\hat{\Pi}^{\theta^*}$ where:

$$\hat{\pi}_1^{\theta^*}(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq F^{-1}\left((\frac{w_2}{w_1})^{\frac{1}{N-1}}\right) \text{ or if } \theta_1 < \theta^* \\ \frac{1}{\frac{w_1}{w_2} + 1} & \text{if else} \end{cases}$$

$$\hat{\pi}_2^{\theta^*}(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq F^{-1}\left((\frac{w_2}{w_1})^{\frac{1}{N-1}}\right) \\ 0 & \text{if } \theta_2 < \theta^* \\ \frac{1}{\frac{w_1}{w_2} + 1} & \text{if else} \end{cases}$$

As θ^* could take any value that is no greater than $F^{-1}\left((\frac{w_2}{w_1})^{\frac{1}{N-1}}\right)$, this implies that there is a continuum of equilibria that could arise from any arbitrarily small unilateral deviation from the symmetric equilibrium. In the following section, we present a special example of this type of deviation that illustrates how it might arise in real life.

4.2 Ex-ante Heterogeneity

In this section, we present a simple motivating example of ex-ante heterogeneous workers to highlight why we may want to study equilibria of $\Gamma(N, w_1, w_2, F(\cdot))$ other than Π^S.

Consider the game $\Gamma(N, w_1, w_2, \mathbb{F})$, where $F_1(\cdot)$ and $F_2(\cdot)$ are such that $F_2^{-1}(\frac{w_2}{w_1}) \neq F_1^{-1}(\frac{w_2}{w_1})$. We assume W.L.O.G. that $F_2^{-1}(\frac{w_2}{w_1}) < F_1^{-1}(\frac{w_2}{w_1})$. In practice, these differences in distributions could arise from a number of sources. For instance, worker 1 could have graduated from a more prestigious school, or is from an ethnic group traditionally associated with higher skill.

The game has a unique equilibrium $\tilde{\Pi}(\theta) = \begin{bmatrix} \tilde{\pi}_1(\theta_1) \\ \tilde{\pi}_2(\theta_2) \end{bmatrix}$ in which:

$$\tilde{\pi}_1(\theta_1) = 1 \ \ \forall \theta_1 \in [a, b]$$
$$\tilde{\pi}_2(\theta_2) = \begin{cases} 1 & \text{if } \theta_2 \geq F_1^{-1}(\frac{w_2}{w_1}) \\ 0 & \text{if else} \end{cases}$$

We show in the appendix that $\tilde{\Pi}$ is the unique equilibrium of the game.

Intuitively, any worker that is sufficiently skilled finds it optimal to apply to the high paying firm regardless of the school he graduated from or what the other worker is doing. However, as his skill drops, he becomes less confident in his ability to get the job offered by firm 1. There comes a threshold $x_i (x_1 = F_1^{-1}(\frac{w_2}{w_1})$ for worker 2 and $x_2 = F_2^{-1}(\frac{w_2}{w_1})$ for worker 1) where the worker finds it more appealing to apply to firm 2 if he thinks the other worker is always applying to firm 1. In general, when workers are ex-ante heterogeneous, the two thresholds will not be identical. However, if worker i only applies to firm 1 if his skill θ_i is above a certain threshold x_i, then by the downward recursive structure of the payoff function, worker j's payoff from applying to firm 1 is constant for any $\theta_j \leq x_i$. As a result, the worker with the lower threshold finds it optimal to apply to firm 1 even when his realised skill is extremely low because he knows the other worker would only apply to firm 1 with a high skill realisation. In some sense, we can think of the worker with the lower threshold as being more confident due to his education or ethnic background. This confidence then magnifies the ex-ante heterogeneity into drastically different application strategies.

Another way of understanding this equilibrium is in the context of a correlated equilibrium. In this interpretation, the difference in distributions can be viewed as a correlation device that allows the workers to coordinate on their application strategies. This interpretation is particularly appealing when considering the case when the differences between $F_1(\cdot)$ and $F_2(\cdot)$ are arbitrarily small and are little more than publicly observable signals.

When $F_1(\cdot)$ and $F_2(\cdot)$ are arbitrarily close to each other but $F_2^{-1}(\frac{w_2}{w_1}) < F_1^{-1}(\frac{w_2}{w_1})$, the overall equilibrium structure would remain the same. This implies that arbitrarily small differences in the skill distributions are sufficient to create large asymmetry in the equilibrium outcome. Formally, let $F(\cdot)$ be the CDF of a continuous distribution of [a,b]. Then as:

$$\max_{i=1,2}\{\sup_{x\in[a,b]}\{|F_i(x)-F(x)|\}\}\to 0$$

The equilibrium strategy profile $\tilde{\Pi}$ converges to:

$$\tilde{\pi}_1(\theta_1)=1 \ \ \forall\theta_1\in[a,b]$$
$$\tilde{\pi}_2(\theta_2)=\begin{cases}1 & \text{if } \theta_2\geq F^{-1}(\frac{w_2}{w_1})\\ 0 & \text{if else}\end{cases}$$

Which corresponds to $\hat{\Pi}^{\theta^*}$ when $\theta^*=F^{-1}\left(\frac{w_2}{w_1}\right)$. We denote this limit equilibrium by $\tilde{\Pi}^A$. Because workers' skill distributions are almost certainly not exactly identical in real life, it could be argued that $\tilde{\Pi}^A$ is more likely to be an accurate description of real life human behaviours than Π^S. As we will show in the following section, $\tilde{\Pi}^A$ in fact belongs to a class of equilibria that is generally robust to any minor ex-ante heterogeneity.

4.3 Asymmetric Equilibrium

In this section, we generalise the $\tilde{\Pi}^A$ equilibrium profile proposed in the previous one, and discuss some of its properties. We first introduce a general characterisation of equilibrium profiles that are in the same spirit as $\tilde{\Pi}^A$.

Proposition 2: For any continuously differentiable distribution $F(\cdot)$ with support on [a,b], the game $\Gamma(N,w_1,w_2,F(\cdot))$ has an asymmetric Nash equilibrium $\Pi^A(\theta)=\begin{bmatrix}\pi_1^A(\theta_1)\\ \dots\\ \pi_N^A(\theta_N)\end{bmatrix}$ where:

$$\pi_i^A(\theta_i)=\begin{cases}1 & \textit{if } \theta_i\geq x_{J(i)}\\ 0 & \textit{if else}\end{cases}$$

Where $\{(x_j,k_j)\}_{j=1,\dots,J}$[2] are defined by the following recursive algorithm:

1. $x_1=F^{-1}\left((\frac{w_2}{w_1})^{\frac{1}{N-1}}\right)$, $k_1=\lceil\frac{N-1}{1+F(x_1)}\rceil$

[2]Throughout this paper, we assume away situations in which: $\frac{1}{1+F(x_j)}(N-1-\sum_{h=1}^{j-1}k_h\frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)})$ is an exact integer.

2. *If* $j > 1$ *and* $\sum_{h=1}^{j-1} k_h < N-1$*:*

$$x_j = \max_{\tilde{x}} \{\tilde{x} < x_{j-1} | w_1 F(\tilde{x})^{N-1-\sum_{h=1}^{j-1} k_h} \prod_{h=1}^{j-1} F(x_h)^{k_h} = w_2 \prod_{h=1}^{j-1} (F(\tilde{x}) + 1 - F(x_h))^{k_h}\}$$

$$k_j = \lceil \frac{1}{1+F(x_j)} (N-1-\sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)}) \rceil$$

3. *If* $j > 1$ *and* $\sum_{h=1}^{j-1} k_h = N-1$*, then define:*

$$\bar{J} = j$$
$$x_j = a$$
$$k_j = 1$$

and terminate the algorithm

Where $\lceil x \rceil$ *is the smallest integer strictly greater than x.* $J(i) : \{1,...,N\} \to \{1,...,\bar{J}\}$ *is a function assigning agent i into one of* $\bar{J}$ *groups:*

$$J(i) = \min_{h \in \mathbb{N}^+} \{h | \sum_{j=1}^{h} k_j \geq i\}$$

It can be easily shown that when $N = 2$, Π^A reduces to $\tilde{\Pi}^A$ from the preceding section. Intuitively, Π^A is a finite worker generalisation of $\tilde{\Pi}^A$ that assigns each worker to one of $\bar{J}$ groups. The size of group j is given by k_j. The group size is chosen to ensure all workers would be content with their group assignment. Unlike the symmetric equilibrium, in which all agents play the same strategy, workers within the same group play the same strategy but workers from different groups follow different strategies. Each agent follows a threshold strategy where he applies to firm 1 with certainty if his skill is above a group-specific threshold and applies to firm 2 with certainty if otherwise. Groups differ in their sizes and their thresholds.

The following image provides a graphical illustration of the two equilibrium profiles of the game $\Gamma(6,1,0.5,F(x)=x)$:

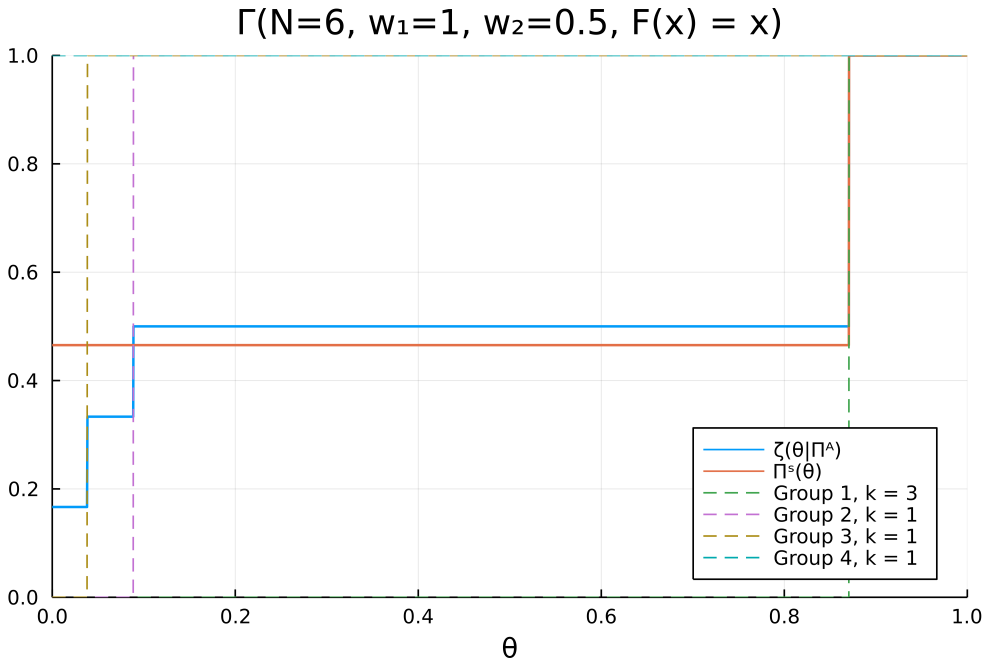


Figure 1: Symmetric and Asymmetric Equilibria

The orange line represents $\pi_i^S(\theta_i)$, and the four dashed lines represent $\pi_i^A(\theta_i)$ for $J(i) = 1, 2, 3,$ and 4 respectively. In this particular game, the asymmetric equilibrium consists of 4 groups of workers, with group 1 having 3 workers and groups 2, 3, and 4 having 1 worker each. Workers in group 1 apply to firm 1 if and only if their skills are above $x_1 = 0.8706$, the workers in groups 2 and 3 apply to firm 1 if and only if their skills are above $x_2 = 0.0887$ and $x_3 = 0.0383$ respectively, and the worker in group 4 always applies to firm 1. In general, there will always be exactly one worker who always applies to firm 1 regardless of his or her skill realisation.

The blue line - $\zeta(\theta|\Pi^A) = \frac{1}{6}\sum_{i=1}^{6} \pi_i^A(\theta)$ - represents the aggregate behaviour associated with Π^A. It is calculated by taking the average of $\pi_i^A(\theta)$ across all i, and represents the probability that a randomly selected worker of a given skill level applies to firm 1. As can be seen from the graph, the aggregate behaviours between the two equilibria are exactly identical for $\theta > x_1$, and are fairly similar for $\theta \in (x_2, x_1)$. However, the difference between the equilibria becomes more pronounced for workers with lower skill realisations.

Much like Π^S, Π^A is monotonic by construction. This assures that workers with higher skills are more likely to be hired by the higher paying firm. While this has no welfare implication in our model, it generally implies higher total welfare in richer models with an active production process where total output depends on the productivities of both the firm and worker.

By the characterisations of the equilibria, aggregate application decisions are always identical across Π^A and Π^S for any $\theta > x_1$. It can also be shown fairly easily that as N goes to infinity, the difference between aggregate application decisions goes to zero at n^{-1} rate for any θ that lies between the thresholds of the first and second group. However, for θ that are smaller than the sec-

ond threshold, the plot shows significant differences in aggregate behaviours, and the difference increases as θ becomes lower. While it is generally true that the difference between the equilibria decreases as N grows, it is unclear if the difference goes to zero everywhere as N goes to infinity.

4.3.1 Robustness

As mentioned in Section 4.2, an important reason to study alternate equilibrium characterisations is because Π^S is very sensitive to minute ex-ante heterogeneity. Much like $\tilde{\Pi}^A$, Π^A is also robust to minute heterogeneity in skill distributions. In this section, we formally establish the robustness of Π^A:

We first introduce the following distance definition for vectors of functions and the concept of equivalence under reordering:

Definition 2: We say one strategy profile $\tilde{\Pi}$ is a reordering of another strategy profile $\bar{\Pi}$ if and only if there exists a bijective function $B(i) : \{1,...,N\} \to \{1,...,N\}$ such that:

$$\sup_{x\in[a,b]} |\tilde{\pi}_i(x) - \bar{\pi}_{B(i)}(x)| = 0 \ \ \forall i \in \{1,...,N\}$$

Definition 3: Let $\tilde{F} = \begin{bmatrix} \tilde{F}_1(\cdot) \\ ... \\ \tilde{F}_N(\cdot) \end{bmatrix}$ and $\hat{F} = \begin{bmatrix} \hat{F}_1(\cdot) \\ ... \\ \hat{F}_N(\cdot) \end{bmatrix}$ be two vectors of functions defined over intervals $[a,b]$. We define the distance between $\tilde{F}$ and $\hat{F}$ as:

$$d(\tilde{F},\hat{F}) = \max_{i=1,...,N} \left\{ \sup_{x\in[a,b]} \{|\hat{F}_i(x) - \tilde{F}_i(x)|\} \right\}$$

We now define the notion of robustness:

Definition 4: Let $F(\cdot)$ be the CDF of a distribution on [a,b] that is continuously differentiable everywhere. Let $F = \begin{bmatrix} F(\cdot) \\ ... \\ F(\cdot) \end{bmatrix}$ be a vector of length N whose entries are all equal to $F(\cdot)$. Let $\mathbb{F}^m = \begin{bmatrix} F_1^m(\cdot) \\ ... \\ F_N^m(\cdot) \end{bmatrix}$ be an arbitrary vector of distributions such that $F_i^m(\cdot)$ are all continuously differentiable for every i and m and has support [a,b]. Let $f = \begin{bmatrix} f(\cdot) \\ ... \\ f(\cdot) \end{bmatrix}$, and $f^m = \begin{bmatrix} f_1^m(\cdot) \\ ... \\ f_N^m(\cdot) \end{bmatrix}$ be the corresponding vectors of PDFs for F and $\mathbb{F}^m$. Let Π be a Nash equilibrium of $\Gamma(N, w_1, w_2, F(\cdot))$. We say Π is robust to small ex-ante heterogeneity, or a robust equilibrium for short, if for any $\delta > 0$, there exists $\varepsilon > 0$ such that $\forall \mathbb{F}^m$ with $\max\{d(\mathbb{F}^m, F), d(f^m, f)\} < \varepsilon$, there exists Π^m and $\tilde{\Pi}$ where:

- *Π^m is a Nash equilibrium of $\Gamma(N, w_1, w_2, \mathbb{F}^m)$*

- $\tilde{\Pi}$ *is a reordering of* Π
- $d(\Pi^m, \tilde{\Pi}) < \delta$

Intuitively, robustness simply states that the equilibrium behaviours of workers who are exactly identical should be approximately identical to the behaviours of workers who are approximately identical (under some reordering). As can be seen from the example in Section 4.2, Π^S is not a robust equilibrium. The following proposition states that Π^A is robust.

Proposition 3: For any continuously differentiable distribution $F(\cdot)$ *with support [a,b],* $w_1 > w_2$*, and finite N,* Π^A *is a robust equilibrium of the game* $\Gamma(N, w_1, w_2, F(\cdot))$

To see why Proposition 3 might hold, note that Π^A is a pure strategy equilibrium by construction. As a result, for any given skill realisation, workers generally strictly prefer one firm or another in equilibrium. As payoffs are continuous in $\mathbb{F}$, minute changes in $\mathbb{F}$ are unlikely to reverse the preference relation.

Proposition 3 also generalises the insight on how Π^A might arise in practice. As workers' skills are likely drawn from similar but non-identical distributions, workers' application strategies are more likely to be in the neighbourhood of a robust equilibrium. Workers use the ex-ante heterogeneity (education or ethnic background) as a correlation device to sort themselves into various groups of appropriate sizes. While the exact assignment of workers to groups depends on the unobservable vector of skill distributions, aggregate variables such as the number and sizes of groups can be calculated from the easily observable wage profiles and number of workers.

4.3.2 Uniqueness

In this section, we extend on the robustness result to show that not only is Π^A a robust equilibrium, it is also the only (up to some reordering) robust equilibrium. To do so, we first show that all pure strategy equilibria are just reorderings of Π^A. We then show that any equilibrium containing mixed strategies cannot be robust.

Formally, we introduce the following definition of a pure strategy equilibrium and the proposition:

Definition 5: $\tilde{\Pi} = \begin{bmatrix} \tilde{\pi}_1(\cdot) \\ ... \\ \tilde{\pi}_N(\cdot) \end{bmatrix}$ *is a pure strategy profile if and only if* $\forall \theta \in [a,b]^N$, $\tilde{\Pi} \in \{0,1\}^N$.
Or equivalently, $\tilde{\Pi}$ *is a pure strategy profile if and only if:*

$$\pi_i(\theta_i) \in \{0,1\} \;\; \forall i \in \{1,...,N\}, \;\; \forall \theta_i \in [a,b]$$

A pure strategy equilibrium is a Nash equilibrium that is also a pure strategy profile; a mixed strategy equilibrium is a Nash equilibrium that is not a pure strategy profile

Proposition 4: Π^A *is the "unique" pure strategy equilibrium in that any pure strategy equilibrium* Π' *is a reordering of* Π^A.

While the proof of Proposition 4 is fairly long and algebraically dense, the basic intuition is relatively simple. First of all, note that any pure strategy equilibrium features groups of workers switching between applying to firm 1 and 2 when their skills are above or below certain thresholds. Therefore, we can characterise a pure strategy equilibrium, up to some reordering, by describing the sizes of each group and their group specific strategies. If a group is using a monotonically increasing strategy ($\pi_i(\theta) \geq \pi_i(\theta') \; \forall \theta > \theta'$), workers of this group apply to firm 1 if their skill is above some constant threshold and apply to firm 2 if their skill is below it. Therefore, the group's strategy can be characterised by a single constant representing the value of this threshold. As it turns out, any pure strategy equilibrium includes only monotonically increasing strategies. This makes some intuitive sense as higher skill should be associated with a greater willingness to go after the high paying but more competitive job. Now consider the group with the highest threshold. The value of the threshold is uniquely determined by the highest skill at which a worker is indifferent between applying to firm 1 and firm 2, which is always unique for any game $\Gamma(N, w_1, w_2, F(\cdot))$. Furthermore, if the group size is too large, then competition for firm 2 would be too stiff for a group member whose skill is just below the threshold. Such a member would then have an incentive to exit the group and adopt a lower threshold. Similarly, if the group size is too small, then competition for firm 1 would be too stiff for a non-group member whose skill is just below the threshold, and he would want to join group 1 and adopt that group's threshold. As a result, both the size of the group and the value of the threshold are uniquely determined by equilibrium conditions.

Having shown that all pure strategy Nash equilibria are reorderings of Π^A, we next introduce the following proposition which provides a sufficient condition under which mixed equilibria are non-robust.

Proposition 5: Let $F(\cdot)$ *be the CDF of a distribution on [a,b] that is continuously differentiable everywhere. Let* $F = \begin{bmatrix} F(\cdot) \\ \dots \\ F(\cdot) \end{bmatrix}$ *be a vector of length N whose entries are all equal to* $F(\cdot)$. *Let* $\mathbb{F}^m = \begin{bmatrix} F_1^m(\cdot) \\ \dots \\ F_N^m(\cdot) \end{bmatrix}$ *be an arbitrary vector of distributions such that* $F_i^m(\cdot)$ *are all continuously differentiable for every i and m and has support [a,b]. Let* $f = \begin{bmatrix} f(\cdot) \\ \dots \\ f(\cdot) \end{bmatrix}$, *and* $f^m = \begin{bmatrix} f_1^m(\cdot) \\ \dots \\ f_N^m(\cdot) \end{bmatrix}$ *be the corresponding vectors of PDFs for* F *and* $\mathbb{F}^m$. *Let* $\mathbb{S}^m$ *be the set of Nash equilibria of the game* $\Gamma(N, w_1, w_2, \mathbb{F}^m)$. *For each* $\tilde{\Pi}^m \in \mathbb{S}^m$, *define* $\tilde{x}_i^m$ *as:*

$$\tilde{x}_i^m = \inf\{x | \tilde{\pi}_i^m(\theta) = 1 \; \forall \theta > x\}$$

Then $\exists \varepsilon > 0$ *such that for any* F^m *satisfying* $d(F^m, F) < \varepsilon$, *if* $\exists \Pi^m \in \mathbb{S}^m$ *such that* $x_i^m \neq x_j^m \; \forall i \neq j$, *then every strategy profile in* $\mathbb{S}^m$ *is a reordering of* Π^m. *Furthermore,* $\Pi^m = \begin{bmatrix} \pi_1^m(\cdot) \\ \dots \\ \pi_N^m(\cdot) \end{bmatrix}$ *is such that:*

$$\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if otherwise} \end{cases}.$$

Proposition 5 is fairly abstract. Intuitively, in equilibrium, all agents with sufficiently high skills want to apply to firm 1. However, as their skills drop, some agents will start applying to firm 2 with positive probability. When agents' skills follow different distributions, then different agents may have different switching points (x_i^m) based on their skill distributions. The proposition states that if agents are sufficiently similar to each other, then in any equilibrium where all agents have different switching points from each other, any agent whose skill is below his switching point will find it strictly optimal to apply to firm 2.

In general, it is difficult to characterise agents' equilibrium behaviour at and below their switching points. A worker could choose to apply to both firms with positive probability when his skill falls below his switching point; or he could apply to firm 2 with certainty if his skill is only slightly below his switching point, but go back to applying to firm 1 if his skill drops even lower. Proposition 5 outlines a sufficient condition under which both of these cases are ruled out.

Importantly, Proposition 5 implies that any equilibrium profile of the game $\Gamma(N, w_1, w_2, F(\cdot))$ that contains mixed strategies cannot be robust if a vector of distributions exists in the neighbourhood of $F(\cdot)$ such that the conditions of Proposition 5 hold. The following lemma states that such a vector of distributions can always be found.

Lemma 1: Let $\mathbb{F}^m = \begin{bmatrix} F_1^m(\cdot) \\ ... \\ F_N^m(\cdot) \end{bmatrix}$ be an arbitrary vector of distributions such that $F_i^m(\cdot)$ are all continuously differentiable for every i and m and has support [a,b]. Let $\mathbb{S}^m$ be the set of Nash equilibria of the game $\Gamma(N, w_1, w_2, \mathbb{F}^m)$. For each $\tilde{\Pi}^m \in \mathbb{S}^m$, define $\tilde{x}_i^m$ as:

$$\tilde{x}_i^m = \inf\{x | \tilde{\pi}_i^m(\theta) = 1 \ \forall \theta > x\}$$

If $F_i^m(\cdot)$ first-order stochastically dominates $F_{i-1}^m(\cdot)$ $\forall i \in \{2, ..., N\}$, then for any $\tilde{\Pi}^m \in \mathbb{S}^m$:

$$\tilde{x}_{i-1}^m > \tilde{x}_i^m \ \ \forall i \in \{2, ..., N\}$$

Intuitively, if workers' skill distributions are strictly ranked by first-order stochastic dominance, then workers can use that ranking to organise themselves into an ordering. The worker with the worst skill distribution has a very high threshold because he knows that the likelihood of his opponents having higher skills than him is high. As a result, he only applies to firm 1 when he himself is extremely skilled. Meanwhile, the worker with the best skill distribution knows that his opponents are more likely to be low skilled, and is willing to apply to firm 1 even when his own skill realisation is extremely poor.

Combined with Proposition 5, Lemma 1 implies that any mixed strategy equilibrium of $\Gamma(N, w_1, w_2, F(\cdot))$ cannot be robust.

Proposition 6: Let $\Pi^S = \begin{bmatrix} \pi_1^S(\cdot) \\ ... \\ \pi_N^S(\cdot) \end{bmatrix}$ *be an equilibrium of* $\Gamma(N, w_1, w_2, F(\cdot))$*. If* $\exists i \in \{1, ..., N\}$ *and interval* $[c,d] \subset [a,b]$ *such that* $\pi_i^S(\theta_i) \in (0,1)\ \forall \theta_i \in [c,d]$*, then* Π^S *cannot be a robust equilibrium.*

Combining Propositions 4 and 6, we obtain the main result of the paper:

Theorem 1: An equilibrium $\tilde{\Pi}$ *of* $\Gamma(N, w_1, w_2, F(\cdot))$ *is robust if and only if it is a reordering of* Π^A

Because all robust equilibria are reorderings of each other, all robust equilibria of a game $\Gamma(N, w_1, w_2, F(\cdot))$ possess the same aggregate properties such as average application decision or likelihood of congestion. This makes robustness a meaningful equilibrium refinement criterion in addition to being a desirable property.

4.3.3 Group Assignment

In addition to being part of the proof for the uniqueness of the robust equilibrium, Lemma 1 also has interesting implications regarding group assignment. From the definition of robustness, any asymmetric game is guaranteed to have an equilibrium "near" one of the robust equilibria of the symmetric game. However, given an arbitrary vector of distributions $\mathbb{F}$ that is "close" to some distribution $F(\cdot)$, it is in general difficult to analytically determine which robust equilibrium of $\Gamma(F(\cdot))$ best approximates an equilibrium of the asymmetric game $\Gamma(\mathbb{F})$.

When all elements of $\mathbb{F}$ can be ranked by first-order stochastic dominance and are sufficiently close to $F(\cdot)$, Lemma 1 states that there is a unique equilibrium where all workers use a pure threshold strategy with the ranking of the threshold given by the first-order stochastic dominance ranking of their distributions. Workers whose skill distributions are dominated have higher thresholds in equilibrium. As a result, the robust equilibrium of $\Gamma(F(\cdot))$ that is closest to the equilibrium of $\Gamma(\mathbb{F})$ is the one that assigns workers with dominated skill distributions to groups with higher thresholds.

This result is relevant in applications where the analyst:

- Believes that workers have different skill distributions, and that the skill distributions are ranked by first-order stochastic dominance
- Believes the workers agree on the relative ranking of each other's distributions, even if they do not exactly know each others' distributions
- Knows the ranking of the workers' skill distributions, i.e., which worker has the best distribution, which worker has the second best, etc.
- Does not know the entire vector of skill distributions, but knows that all distributions are fairly similar to some distribution $F(\cdot)$

In such a setting, Lemma 1 pins down the unique robust equilibrium that would best approximate the equilibrium behaviour. One application that satisfies the above condition is labour markets with racial or gender discrimination. In such a setting, the model would predict that a member of disadvantaged groups would adopt a more conservative application strategy where he or she only applies to the high paying job if the skill realisation is high; while members of the more privileged groups would adopt a highly aggressive application strategy that sees them going after high paying jobs even when their skills are extremely low.

5 Large Markets

As Π^A does not have an analytical solution in general, it is difficult to determine its properties beyond the ones implied by its characterisation. Focusing on the limiting behaviour of Π^A as the number of workers goes to infinity would somewhat circumvent this challenge, but it is conceptually problematic to think of an economy with an arbitrarily large number of workers but only two job openings. In this section, we modify the framework to allow the number of firms offering each wage to grow with the number of workers while restricting the number of wage levels to two, and study the behaviour of the robust equilibrium in a "large" economy.

5.1 Definition

Formally, we define the following replicator economy. As before, let N denote the total number of workers. We assume there are two classes of firms, one offering wage w_1 and the other offering wage w_2. For the sake of brevity, we still refer to any firms offering $w_{1(2)}$ as firm 1(2). Let H_1 and H_2 denote the number of firms offering wages w_1 and w_2 respectively. We assume that $\frac{H_1}{N} \to \alpha_1$, $\frac{H_2}{N} \to \alpha_2$, where $\alpha_1, \alpha_2 > 0$, and $\alpha_1 + \alpha_2 < 1$. For the sake of convenience, we assume that N is such that $H_1 = \alpha_1 N$ and $H_2 = \alpha_2 N$ are always integers. For the remainder of this analysis, we also normalise w_1 to 1 and assume $w_2 < 1$. Formally, we denote a game with N workers, $H_1 = \alpha_1 N$ firms offering $w_1 = 1$, $H_2 = \alpha_2 N$ firms offering $w_2 < 1$, and workers whose skills are distributed according to a distribution with CDF $F(\cdot)$ as $\Gamma^R(N, w_2, \alpha_1, \alpha_2, F(\cdot))$

Just like before, we assume that workers have perfect private information about their own skill realisations, and formulate their application decisions based on their own skills. However, as workers are indifferent between any firms offering the same wage, we assume that workers choose the class of firm they wish to apply to, and then randomly select one of the firms in that class. Similar to before, we denote worker i's action by a_i, where $a_i = 1$ if he decides to apply to firm 1, and $a_i = 2$ if he applies to firm 2. Given a strategy profile Π, a worker i's payoff given his skill realisation θ_i is given by:

$$u_i^R(a_i|\theta_i, \Pi_{-i}) = \begin{cases} \prod_{j=1, j\neq i}^{N} \left(1 - \frac{\int_{\theta_i}^{b} \pi_i(\mu) f(\mu) \partial\mu}{H_1}\right) & \text{if } a_i = 1 \\ w_2 \prod_{j=1, j\neq i}^{N} \left(1 - \frac{\int_{\theta_i}^{b} \left(1 - \pi_j(\mu)\right) f(\mu) \partial\mu}{H_2}\right) & \text{if } a_i = 2 \end{cases}$$

The payoffs are very similar to the ones in the original game, with the only noticeable difference being that the probability a worker gets outcompeted by another worker is divided by the number of firms in the class to account for the fact that workers applying to the same class of firms need not be applying to the same firm. Due to this overall similarity, most of the earlier results carry over with some minimal modifications.

5.2 Equilibrium

The following proposition defines the robust equilibrium for $\Gamma^R(N, w_2, \alpha_1, \alpha_2, F(\cdot))$.

Proposition 7: The game $\Gamma^R(N, w_2, \alpha_1, \alpha_2, F(\cdot))$ *has a robust Nash equilibrium* $\Pi^R(\theta) = \begin{bmatrix} \pi_1^R(\theta_1) \\ \dots \\ \pi_N^R(\theta_N) \end{bmatrix}$ *where:*

$$\pi_i^R(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_{J^R(i)} \\ 0 & \text{if else} \end{cases}$$

Where $\{(x_j, k_j)\}_{j=1,\dots,\bar{J}}$ *is defined as:*

1. $x_1 = \begin{cases} F^{-1}\left(1 - H_1[1 - w_2{}^{\frac{1}{N-1}}]\right) & \text{if } H_1[1 - w_2{}^{\frac{1}{N-1}}] < 1 \\ a & \text{if else} \end{cases}$, $k_1 = \begin{cases} \lceil \frac{N-1}{1+G(x_1)} \rceil & \text{if } x_1 > a \\ k_1 = 0 & \text{if } x_1 = a \end{cases}$

2. *If* $j > 1$ *and* $x_{j-1} \neq a$*:*

$$x_j = \max\left\{ \{a\} \cup \{\tilde{x} < x_{j-1} | G(\tilde{x})^{N-1-\sum_{h=1}^{j-1} k_h} \prod_{h=1}^{j-1} G(x_h)^{k_h} = w_2 \prod_{h=1}^{j-1} (G(\tilde{x}) + 1 - G(x_h))^{k_h} \} \right\}$$

$$k_j = \begin{cases} \lceil \frac{1}{1+G(x_j)} (N - 1 - \sum_{h=1}^{j-1} k_h \frac{2G(x_j)+1-G(x_h)}{G(x_j)+1-G(x_h)}) \rceil & \text{if } x_j = a \\ 0 & \text{if else} \end{cases}$$

3. *If* $j > 1$ *and* $x_{j-1} = a$*, then define* $\bar{J} = j - 1$ *and terminate the algorithm*

Where $G(\theta) = \frac{H_1 - 1}{H_2} + \frac{1}{H_2} F(\theta)$, $J^R(i) = \min_{h \in \mathbb{N}^+} \{h | \sum_{j=1}^{h} k_j \geq i\}$. *Furthermore,* Π^R *is the unique robust equilibrium up to reordering.*

Similar to Π^A, Π^R is only defined recursively and lacks a closed form solution. As a result, it is difficult to analytically examine the properties of Π^R. From the construction algorithm, we can infer that, much like Π^A, Π^R is a pure strategy equilibrium that only features monotonic strategies. Workers are still assigned to groups that play group specific threshold strategies. The similarity between Π^A and Π^R comes from the similarity in workers' payoffs in Γ and Γ^R. It can be shown that the payoffs under Γ^R can be expressed in the same form as the payoffs under Γ with some creative algebra and change of variables.

Most of the difference between Π^A and Π^R comes from the fact that

$$G(\tilde{x})^{N-1-\sum_{h=1}^{j-1}k_h}\prod_{h=1}^{j-1}G(x_h)^{k_h} = w_2\prod_{h=1}^{j-1}(G(\tilde{x})+1-G(x_h))^{k_h}$$

may not have any solution in the $[a,b]$ interval. Mathematically, this is because $G(a) = \frac{H_1-1}{H_2} > 0$, and the intermediate value theorem no longer applies. Intuitively, in the replicator economy, a worker with a skill realisation of a may still be hired by firm 1 even if there are other workers who always apply to firm 1. As a result, it is possible to have more than one worker who always applies to firm 1. Similarly, it is also possible to have all workers apply to firm 1 regardless of their skill realisations if w_2 is too low or if α_1 is too large. For the remainder of this paper, we focus on the more interesting case of when $w_2 > e^{-\frac{1}{\alpha_1}}$ - i.e. when some workers find it optimal to switch to firm 2 when their skills are too low.

5.2.1 Aggregate Application Decision

In this section, we study the aggregate application decision of workers in a replicator economy as $N \to \infty$. Aggregate behaviour is important in determining a wide array of phenomena studied under the directed search framework, including but not limited to congestion (when more than two workers apply to the same firm) and assortative matching. Formally, we define the aggregate application decision induced by a strategy profile Π as:

$$\zeta(\theta|\Pi) = \text{Prob}(a_i = 1|\theta) = \frac{1}{N}\sum_{i=1}^{N}\pi_i(\theta)$$

For the sake of notational convenience, we also define $\zeta^N(\theta) = \zeta(\theta|\Pi^R)$ as the aggregate application decision implied by a robust equilibrium of the $\Gamma^R(N, w_2, \alpha_1, \alpha_2, F(\cdot))$. Unlike the individual behaviours, the aggregate behaviour has very tractable asymptotic properties:

Proposition 8: Let Π^R be a robust equilibrium of $\Gamma^R(N, w_2, \alpha_1, \alpha_2, F(\cdot))$, and define $\zeta^N(\theta) : [a,b] \to [0,1]$ as $\zeta^N(\theta) = \frac{1}{N}\sum_{i=1}^{N}\pi_i^R(\theta)$. Then for all $\theta \in [a,b]$, if $w_2 > e^{-\frac{1}{\alpha_1}}$, then:

$$\zeta^N(\theta) \to \zeta^*(\theta) = \begin{cases} 1 & \text{if } \theta > x_1 = F^{-1}\left(1+\alpha_1\ln(w_2)\right) \\ \frac{\alpha_1}{\alpha_1+\alpha_2} & \text{if else} \end{cases}$$

if $w_2 \leq e^{-\frac{1}{\alpha_1}}$, then:

$$\zeta^N(\theta) \to 1$$

When w_2 is too small relative to the ratio of firms offering w_1, no workers would ever find it optimal to apply to a firm offering w_2 in the limit. This results in a trivial equilibrium where $\pi_i(\theta_i) = 1\ \forall i$. When w_2 is large relative to the ratio of firms offering w_1, all workers apply to firms offering $w_1 = 1$ if their skill is above a certain threshold, but only a constant fraction of workers applies to w_1 if their skill is below the threshold. Intriguingly, the ratio of workers applying to w_1

is identical to the ratio of firms offering w_1, and independent of the wage profile. Instead, w_2 enters the equation through the threshold x_1.

This result is very similar to the one obtained in Peters (2010), which focused on the symmetric equilibrium. In particular, it can be shown that $\zeta^*(\theta)$ is also the limiting function of the aggregate application decision of the symmetric equilibrium[3]. This provides some reassurance for empirical works based on the symmetric equilibrium, suggesting that, any bias from misspecification vanishes in the limit as long as the analyst's focus is on aggregate rather than individual behaviours.

Finally, it is worth noting that Proposition 8 does not imply that there are only two groups of workers in the limit, with one group of workers always applying to firm 1, and another only applying to firm 1 if their skill realisations are above x_1. Rather, the proposition only implies that the size of any group whose threshold does not converge to x_1 goes to zero relative to total population as total population goes to infinity. In general, due to the recursive nature of the equilibrium characterisation, it is impossible to derive any theoretical properties on the sizes and thresholds of any group other than the first one.

5.2.2 Individual Payoffs

A natural question in any asymmetric equilibrium is whether or not the payoffs are asymmetric as well. Given the context of our problem, this could have important implications for racial equality. Socio-economically disadvantaged ethnic groups could and often do find themselves adopting the less aggressive application strategies. When there is only one firm offering each wage level and the number of workers is very small ($N \leq 3$), it can be easily shown that the worker who always applies to firm 1 always receives weakly higher payoff than any other worker. The following graph plots the payoff of various groups as a function of the worker's skill realisation for the game $\Gamma(3,1,0.7,F(x)=x)$. As can be seen from the graph, workers whose skills are above the threshold of group 1 receive the same payoff regardless of which group they are in, but workers whose skills are below the threshold are better off in group 2 than group 1.

[3]The game $\Gamma(N,w_2,\alpha_1,\alpha_2,F(\cdot))$ has a unique symmetric Nash equilibrium in which every agent follows $\pi^S(\theta) = \begin{cases} 1 & \text{if } \theta \geq x_1 \\ \frac{\alpha_1}{\alpha_2(\frac{1}{w_2})^{\frac{1}{N-1}}+\alpha_1} & \text{if not} \end{cases}$

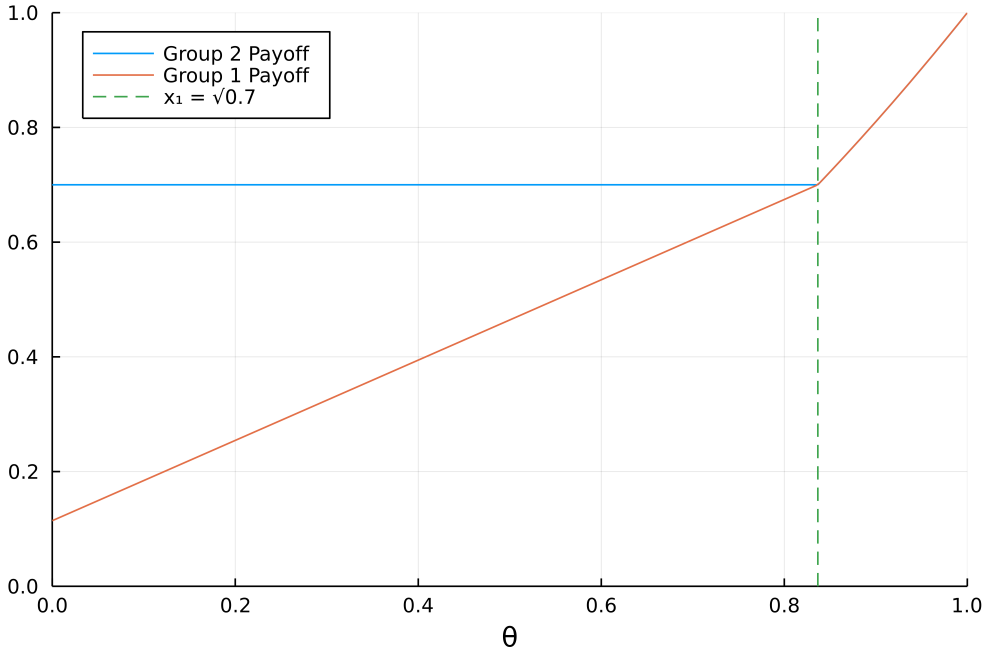


Figure 2:

However, this result does not extend to larger economies. Consider a replicator economy $\Gamma^R(N, w_2, \alpha_1, \alpha_2, F(\cdot))$, a worker i with skill θ_i and a robust equilibrium of the replicator economy $\Pi^{R,N}$. Assume W.L.O.G. that $\theta_i \in [x_{j,N}, x_{j-1,N}]$ for some $j \in \{1, ..., \bar{J}_N - 1\}$ [4]. His equilibrium payoff is given by:

$$u_i(1|\theta_i, \Pi^{R,N}) = \frac{1}{1 - \frac{1-F(\max\{\theta_i, x_{J^R(i),N}\})}{H_1}} \left[\prod_{h=1}^{j-1} (1 - \frac{1-F(x_{h,N})}{H_1})^{k_{h,N}} \prod_{h=j}^{\bar{J}_N} (1 - \frac{1-F(\theta_i)}{H_1})^{k_{h,N}} \right]$$

$$u_i(2|\theta_i, \Pi^{R}) = w_2 \frac{1}{1 - \frac{F(x_{J^R(i),N}) - F(\min\{\theta_i, x_{J^R(i),N}\})}{H_1}} \left[\prod_{h=1}^{j-1} (1 - \frac{F(x_{h,N}) - F(\theta_i)}{H_1})^{k_{h,N}} \right]$$

Notice that the components of $u_i(1|\theta_i, \Pi^R)$ and $u_i(2|\theta_i, \Pi^R)$ in the square bracket are identical across all workers i in all groups. Furthermore, because $H_1 = \alpha_1 N$:

$$\lim_{N\to\infty} \frac{1}{1 - \frac{1-F(\max\{\theta_i, x_{J^R(i)}\})}{H_1}} = \lim_{N\to\infty} \frac{1}{1 - \frac{F(x_{J^R(i)}) - F(\max\{\theta_i, x_{J^R(i)}\})}{H_1}} = 1$$

Therefore, in the limit, all workers with the same skill realisations and application decisions receive the same payoff in the robust equilibrium regardless of which group they are assigned to. Intuitively, as N grows large, all workers face approximately the same strategy profile, and as a result receive the same payoff from the same action given the same skill realisation. This further

[4] We define $x_0 = b$

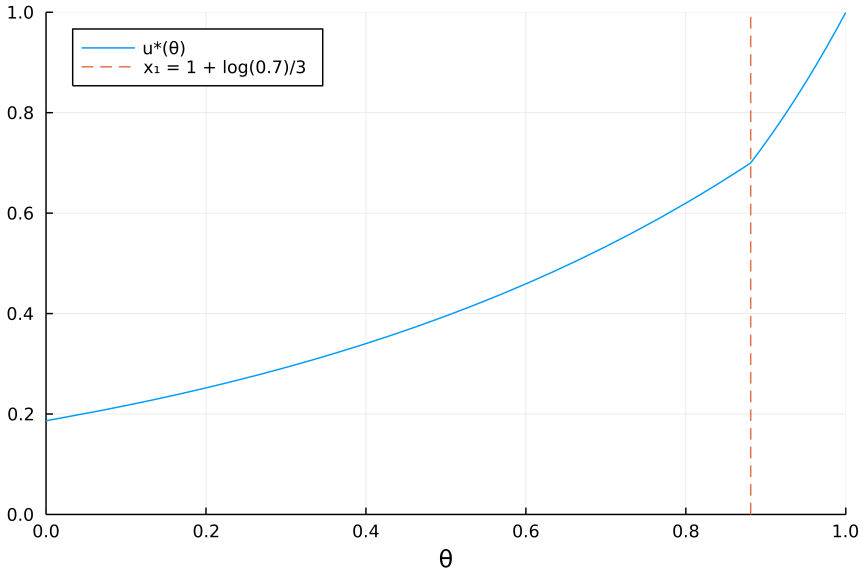


Figure 3:

implies that, in the limit, the payoffs from applying to w_1 and w_2 are identical for all skill realisations below $F^{-1}(1+\alpha_1)$. The following proposition provides a closed form expression for the limiting interim expected payoff of a worker with skill θ_i:

Proposition 9: Let Π^R be the robust equilibrium of the game $\Gamma^R(N, w_2, \alpha_1, \alpha_2, F(\cdot))$, and define worker i's interim payoff given a strategy profile as:

$$u_i^*(\theta|\Pi^R) = u_i(1|\theta_i, \Pi_{-i}^R)\pi_i^R(\theta_i) + u_i(2|\theta_i, \Pi_{-i}^R)[1-\pi_i^R(\theta_i)]$$

Then for any $\theta \in [a,b]$ and any i:

$$\lim_{N\to\infty} u_i^*(\theta|\Pi^R) = u^*(\theta_i) = \begin{cases} (\frac{1}{e})^{\frac{1-F(\theta)}{\alpha_1}} & \text{if } \theta \geq F^{-1}(1+\alpha_1 \ln(w_2)) \\ w_2^{\frac{\alpha_2}{\alpha_1+\alpha_2}} (\frac{1}{e})^{\frac{1-F(\theta)}{\alpha_1+\alpha_2}} & \text{if else} \end{cases}$$

The following graph plots the limiting interim payoff of the replicator economy with $w_2 = 0.7, \alpha_1 = \alpha_2 = \frac{1}{3}, F(x) = x$. Proposition 9 is fairly intuitive. Workers' payoffs are increasing in the wage offered w_2, the number of firms in the economy relative to worker pool α_2 and α_1, and the workers' skill realisation. Intriguingly, the limiting payoff of workers in the robust equilibrium also coincides with the limiting payoff of workers in the symmetric equilibrium[5]. The similarity between the robust equilibrium and the symmetric equilibrium in aggregate variables suggests another interpretation of robustness - the robust equilibrium is a pure strategy approximation of the aggregate dynamics of the mixed strategy equilibria.

[5]See appendix for algebraic details

6 Conclusion

We show that the symmetric equilibrium of a finite directed search game is fragile. Arbitrarily small differences between workers' skill distributions can lead to a unique and sharply asymmetric equilibrium. We characterise the equilibria that survive such differences. An equilibrium is robust to small ex-ante heterogeneity if and only if it is in pure strategies, and this equilibrium exists and is unique up to relabelling the workers. In it, workers sort into groups with distinct skill thresholds. When skill distributions are ranked by first-order stochastic dominance, the ranking orders the thresholds, so workers with dominated distributions apply to the high-wage firm only at high skill.

In large markets the robust and symmetric equilibria differ in individual behaviour but agree in the aggregate. The share of workers at each skill level who apply to the high-wage firm converges to the same limit in both, and so do workers' interim payoffs. Individual application strategies depend sharply on small differences between workers, while aggregate application behaviour does not.

7 Appendix

Uniqueness of the two-worker equilibrium (Section 4.2):

Proof. Consider agent 2's payoff given $\tilde{\pi}_1(\cdot)$:

$$u_2(a_2|\theta_2,\tilde{\pi}_1(\cdot)) = \begin{cases} w_1 F_1(\theta_2) & \text{if } a_2 = 1 \\ w_2 & \text{if } a_2 = 2 \end{cases}$$

Therefore:

$$u_2(1|\theta_2,\tilde{\pi}_1(\cdot)) > u_2(2|\theta_2,\tilde{\pi}_1(\cdot)) \iff \theta_2 > F_1^{-1}(\frac{w_2}{w_1})$$

As a result, given $\tilde{\pi}_1(\cdot)$, it is optimal for agent 2 to apply to firm 1 if and only if $\theta_2 > F_1^{-1}(\frac{w_2}{w_1})$. Now consider agent 1's payoff given $\tilde{\pi}_2(\cdot)$, and suppose $\theta_1 \geq F_1^{-1}(\frac{w_2}{w_1})$:

$$u_1(1|\theta_1,\tilde{\pi}_2(\cdot)) = w_1 F_2(\theta_1) \geq w_1 F_2\left(F_1^{-1}(\frac{w_2}{w_1})\right) > w_1 F_2\left(F_2^{-1}(\frac{w_2}{w_1})\right) = w_2 = u_1(2|\theta_1,\tilde{\pi}_2(\cdot))$$

Now consider agent 1 with $\theta_1 < F_1^{-1}(\frac{w_2}{w_1})$:

$$u_1(1|\theta_1,\tilde{\pi}_2(\cdot)) = w_1 F_2\left(F_1^{-1}(\frac{w_2}{w_1})\right) > w_2 > w_2(F_2(\theta_1) + 1 - F_2\left(F_1^{-1}(\frac{w_2}{w_1})\right)) = u_1(2|\theta_1,\tilde{\pi}_2(\cdot))$$

As a result, it is always strictly optimal for agent 1 to apply to firm 1 regardless of his skill realisations given $\tilde{\pi}_2(\cdot)$.

To show that $\tilde{\Pi}$ is the unique Nash equilibrium, we first note that for any $\pi_2(\cdot) : [a,b] \to [0,1]$:

$$u_1(1|\theta_1, \pi_2) = w_1 \left[F_2(\theta_1) + \int_{\theta_1}^{b} (1 - \pi_2(\mu))\, dF_2(\mu) \right] \geq w_1 F_2(\theta_1)$$
$$u_1(2|\theta_1, \pi_2) = w_2 \left[F_2(\theta_1) + \int_{\theta_1}^{b} \pi_2(\mu) dF_2(\mu) \right] \leq w_2$$

Therefore, for any $\theta_1 > F_2^{-1}(\frac{w_2}{w_1})$, $u_1(1|\theta_1, \pi_2(\cdot)) > u_1(2|\theta_1, \pi_2(\cdot))$ for any arbitrary $\pi_2(\cdot)$. As a result, if worker 1 is playing $\pi_1(\cdot)$ in a Nash equilibrium profile, then $\pi_1(x) = 1 \forall x > F_2^{-1}(\frac{w_2}{w_1})$. By similar logic, we can show that if player 2 is playing $\pi_2(\cdot)$ as part of a Nash equilibrium profile, then $\pi_2(x) = 1 \forall x > F_1^{-1}(\frac{w_2}{w_1})$
Now consider an arbitrary $\pi_1(\cdot)$ and define x^* as $x^* = \inf_{x \in [a,b]} \{x | \pi_1(\theta) = 1 \forall \theta_1 > x\}$. If player 1 is playing $\pi_1(\cdot)$ in a Nash equilibrium, then $x^* \leq F_2^{-1}(\frac{w_2}{w_1}) < F_1^{-1}(\frac{w_2}{w_1})$. Suppose $x^* > a$, then worker 2 with skill $\theta_2 \in \left[x^*, F_1^{-1}(\frac{w_2}{w_1}) \right)$ faces the following payoffs:

$$u_2(2|\theta_2, \pi_1(\cdot)) = w_2 > w_1 F_1(\theta_2) = u_2(1|\theta_2, \pi_1(\cdot))$$

Therefore, if worker 2 is playing $\pi_2(\cdot)$ in any Nash equilibrium where player 1 is playing $\pi_1(\cdot)$, then

$$\pi_2(\theta_2) = \begin{cases} 1 & \text{if } \theta_2 > F_1^{-1}(\frac{w_2}{w_1}) \\ 0 & \text{if } \theta_2 \in \left[x^*, F_1^{-1}(\frac{w_2}{w_1}) \right) \end{cases}$$

However, given such θ_2, a worker 1 with skill $\theta_1 = x^*$ faces payoffs:

$$u_1(1|x^*, \pi_2(\cdot)) = w_1 F_2 \left(F_1^{-1}(\frac{w_2}{w_1}) \right) > w_2 > w_2 \left[F_2(x^*) + 1 - F_2 \left(F_1^{-1}(\frac{w_2}{w_1}) \right) \right] = u_1(2|x^*, \pi_2(\cdot))$$

Because $u_1(a_1|\theta_1, \pi_2(\cdot))$ is continuous in θ_1, there exists some $\varepsilon > 0$ such that:

$$u_1(1|\theta_1, \pi_2(\cdot)) > u_1(2|\theta_1, \pi_2(\cdot)) \ \ \forall \theta_1 \in [x^* - \varepsilon, x^*]$$

As a result, if worker 1 is playing $\pi_1(\cdot)$ in a Nash equilibrium, $\pi_1(\cdot)$ must be such that $\pi_1(\theta_1) = 1$ $\forall \theta_1 \in [x^* - \varepsilon, x^*]$, which violates the definition of x^*. In conclusion, in any Nash equilibrium, player 1 is always applying to firm 1. And as has been shown earlier, if player 1 is always applying to firm 1, then $\tilde{\pi}_2(\cdot)$ is player 2's unique best response. The equilibrium is therefore unique. □

Proposition 2:

Proof. Note that

- $k_j \leq N - 1 - \sum_{h=1}^{j-1} k_h$
- For any j such that $k_{j-1} \neq N - 1 - \sum_{h=1}^{j-2} k_h$, x_j exists. This is because at $\tilde{x} = a$:

$$w_1 F(\tilde{x})^{N-1-\sum_{h=1}^{j-1} k_h} \prod_{h=1}^{j-1} F(x_h)^{k_h} = 0$$

While

$$w_2 \prod_{h=1}^{j-1} (F(\tilde{x}) + 1 - F(x_h))^{k_h} = w_2 \prod_{h=1}^{j-1} (1 - F(x_h))^{k_h} > 0$$

Furthermore, by the inductive proof we shall show below, $\exists \varepsilon > 0$ s.t. $\forall \tilde{x} \in [x_{j-1} - \varepsilon, x_{j-1})$:

$$w_1 F(\tilde{x})^{N-1-\sum_{h=1}^{j-1} k_h} \prod_{h=1}^{j-1} F(x_h)^{k_h} > w_2 \prod_{h=1}^{j-1} (F(\tilde{x}) + 1 - F(x_h))^{k_h}$$

- By the above result, $k_{\bar{J}} = N - 1 - \sum_{h=1}^{\bar{J}-1} k_h$, and $\sum_{h=1}^{\bar{J}} k_h = N - 1$
- Throughout this analysis, we rule out the knife edge cases in which:
  - $\frac{1}{1+F(x_j)}(N - 1 - \sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)})$ is an exact integer
  - x_j is such that the LHS derivative of the following equation with respect to x_j is exactly equal to the RHS derivative of the following equation with respect to x_j:

$$w_1 F(x_j)^{N-1-\sum_{h=1}^{j-1} k_h} \prod_{h=1}^{j-1} F(x_h)^{k_h} = w_2 \prod_{h=1}^{j-1} (F(x_j) + 1 - F(x_h))^{k_h}$$

    In other words, the LHS expression and the RHS expression cross each other at x_j

  These fringe cases do not have any impact on the main results of this paper, but accounting for them requires significantly more cumbersome notations.

Where $\lceil x \rceil$ is the smallest integer strictly greater than x
The proof follows an inductive argument. We first show that for any agent i with skill $\theta_i \geq x_1$, it is optimal for him to follow $\pi_i^A(\theta_i) = 1$ given that everyone else is following Π_{-i}^A.
Next, we show that if it is optimal for any agent i with skill $\theta_i \geq x_j$ to follow $\pi_i^A(\theta_i)$ given that everyone else is following Π_{-i}^A, it is also optimal for any agent i with skill $\theta_i \in [x_{j+1}, x_j)$ to follow $\pi_i^A(\theta_i)$ given that everyone else is following Π_{-i}^A.
We also include some simple arguments on why the equilibrium as defined above is well-defined.

- Base Case:
  Consider an arbitrary agent i with skill $\theta_i \geq x_1$. His payoffs from applying to firm 1 and 2 are given respectively by:

$$u_i(1|\theta_i, \Pi_{-i}^A) = w_1 F(\theta_i)^{N-1}$$
$$u_i(2|\theta_i, \Pi_{-i}^A) = w_2$$

  Note that when $\theta_i = x_1$:

$$u_i(1|x_1, \Pi_{-i}^A) = w_1 F(F^{-1}((\frac{w_2}{w_1})^{\frac{1}{N-1}}))^{N-1} = w_2 = u_i(2|x_1, \Pi_{-i}^A)$$

  and that $u_i(1|\theta_i, \Pi_{-i}^A)$ is increasing in θ_i. Therefore, $\forall \theta_i \geq x_1$:

$$u_i(1|\theta_i, \Pi_{-i}^A) \geq u_i(2|\theta_i, \Pi_{-i}^A)$$

Therefore, for any agent i with skill $\theta_i \geq x_1$, it is optimal for him to follow $\pi_i^A(\theta_i) = 1$ given that everyone else is following Π_{-i}^A.

- Inductive Case:
  Assume that it is optimal for any agent i with skill $\theta_i \geq x_j$ to follow $\pi_i^A(\theta_i)$ given that everyone else is following Π_{-i}^A.
  For the inductive step, we must consider two cases:
  - Consider agent i with $\theta_i \in [x_{j+1}, x_j]$ where $\pi_i^A(\theta_i) = 1$. His payoffs are:

$$u_i^1(1|\theta_i, \Pi_{-i}^A) = w_1 F(\theta_i)^{N-1-\sum_{h=1}^{j} k_h} \prod_{h=1}^{j} F(x_h)^{k_h}$$

$$u_i^1(2|\theta_i, \Pi_{-i}^A) = w_2 \prod_{h=1}^{j} (F(\theta_i) + 1 - F(x_h))^{k_h}$$

    We want to show that $\forall \theta_i \in [x_{j+1}, x_j]$:

$$u_i^1(1|\theta_i, \Pi_{-i}^A) \geq u_i^1(2|\theta_i, \Pi_{-i}^A)$$

    Note that by definition of x_j:

$$u_i^1(1|x_j, \Pi_{-i}^A) = w_1 F(x_j)^{N-1-\sum_{h=1}^{j-1} k_h} \prod_{h=1}^{j-1} F(x_h)^{k_h} = w_2 \prod_{h=1}^{j-1} (F(x_j) + 1 - F(x_h))^{k_h} = u_i^1(2|x_j, \Pi_{-i}^A)$$

    Furthermore, define:

$$H^1(\theta_i) = \ln\left(\frac{u_i^1(1|\theta_i, \Pi_{-i}^A)}{u_i^1(2|\theta_i, \Pi_{-i}^A)}\right)$$

    Note that $H^1(\cdot)$ is continuously differentiable by the continuous differentiability assumption of $F(\cdot)$. Take the first order derivative with respect to θ_i:

$$\frac{\partial H^1(\theta_i)}{\partial \theta_i} = f(\theta_i)\left[\frac{N-1-\sum_{h=1}^{j} k_h}{F(\theta_i)} - \sum_{h=1}^{j} \frac{k_h}{F(\theta_i) + 1 - F(x_h)}\right]$$

    Evaluating the above equation at $\theta_i = x_j$, we obtain:

$$\frac{\partial H^1(\theta_i)}{\partial \theta_i}|_{\theta_i = x_j} = f(x_j)\left[\frac{N-1-\sum_{h=1}^{j} k_h}{F(x_j)} - \sum_{h=1}^{j-1} \frac{k_h}{F(x_j) + 1 - F(x_h)}\right]$$

    Which is greater than zero if and only if:

$$k_j < \frac{1}{1+F(x_j)}\left(N - 1 - \sum_{h=1}^{j-1} k_h \frac{2F(x_j) + 1 - F(x_h)}{F(x_j) + 1 - F(x_h)}\right)$$

    Because

$$k_j = \left\lceil \frac{1}{1+F(x_j)}\left(N - 1 - \sum_{h=1}^{j-1} k_h \frac{2F(x_j) + 1 - F(x_h)}{F(x_j) + 1 - F(x_h)}\right) \right\rceil$$

Therefore

$$k_j > \frac{1}{1+F(x_j)}(N-1-\sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)})$$

and

$$\frac{\partial H^1(\theta_i)}{\partial \theta_i}|_{\theta_i=x_j} < 0$$

Since $H^1(x_j)=0$ by the definition of x_j, $\exists \varepsilon > 0$ such that $\forall \theta_i \in [x_j-\varepsilon, x_j]$:

$$H^1(\theta_i) \geq 0 \iff u_i^1(1|\theta_i, \Pi_{-i}^A) \geq u_i^1(2|\theta_i, \Pi_{-i}^A)$$

Finally, note that by definition of x_{j+1}, $\nexists \theta^* \in (x_{j+1}, x_j)$ s.t.

$$u_i^1(1|\theta_i, \Pi_{-i}^A) = u_i^1(2|\theta_i, \Pi_{-i}^A)$$

Therefore, $\forall \theta_i \in [x_{j+1}, x_j]$:

$$u_i^1(1|\theta_i, \Pi_{-i}^A) \geq u_i^1(2|\theta_i, \Pi_{-i}^A)$$

And the inequality holds strictly if $\theta_i \notin \{x_{j+1}, x_j\}$.

Additionally, note that $k_j \geq 0$. This is because $\frac{\partial H^1(\theta_i)}{\partial \theta_i}$ is decreasing in k_j. Furthermore,

$$\frac{\partial H^1(\theta_i)}{\partial \theta_i}|_{\theta_i=x_j, k_j=0}$$

is greater than or equal to 0 by the definition of x_j (since $H^1(\theta_i|k_j=0) > 0 \ \forall \theta_i \in (x_j, x_{j-1})$ by assumption).

We also know that k_j is such that $\frac{\partial H^1(\theta_i)}{\partial \theta_i}|_{\theta_i=x_j}$ is negative, meaning that k_j must be weakly greater than zero.

– Consider agent i with $\theta_i \in [x_{j+1}, x_j)$ where $\pi_i^A(\theta_i) = 0$. Recycling earlier notations, agent i faces payoffs:

$$u_i^2(1|\theta_i, \Pi_{-i}^A) = w_1 F(\theta_i)^{N-\Sigma_{h=1}^j k_h} \frac{1}{F(x_{J(i)})} \prod_{h=1}^{j} F(x_h)^{k_h}$$

$$u_i^2(2|\theta_i, \Pi_{-i}^A) = w_2 \frac{1}{F(\theta_i)+1-F(x_{J(i)})} \prod_{h=1}^{j} (F(\theta_i)+1-F(x_h))^{k_h}$$

We want to show that $\forall \theta_i \in [x_{j+1}, x_j]$:

$$u_i^2(1|\theta_i, \Pi_{-i}^A) \leq u_i^2(2|\theta_i, \Pi_{-i}^A)$$

Notice that $u_i^2(1|\theta_i, \Pi_{-i}^A)$ is decreasing in $F(x_{J(i)})$ while $u_i^2(2|\theta_i, \Pi_{-i}^A)$ is increasing in $F(x_{J(i)})$. Furthermore, by the definition of x_j:

$$x_1 > x_2 > ... > x_j > x_{j+1} > ...$$

Therefore, if the inequality holds for $J(i) = j$, then it holds for all $J(i) < j$. We can therefore assume W.L.O.G. that i is such that $J(i) = j$, and rewrite the payoffs accordingly:

$$u_i^2(1|\theta_i, \Pi_{-i}^A) = w_1 F(\theta_i)^{N-\sum_{h=1}^{j} k_h} \frac{1}{F(x_j)} \prod_{h=1}^{j} F(x_h)^{k_h}$$

$$u_i^2(2|\theta_i, \Pi_{-i}^A) = w_2 \frac{1}{F(\theta_i)+1-F(x_j)} \prod_{h=1}^{j} (F(\theta_i)+1-F(x_h))^{k_h}$$

Similarly, by the definition of x_j:

$$u_i^2(1|x_j, \Pi_{-i}^A) = u_i^2(2|x_j, \Pi_{-i}^A)$$

Recycling earlier notations, define:

$$H^2(\theta_i) = \ln\left(\frac{u_i^2(1|\theta_i, \Pi_{-i}^A)}{u_i^2(2|\theta_i, \Pi_{-i}^A)}\right)$$

Note that $H^2(\cdot)$ is also continuously differentiable by the continuous differentiability assumption of $F(\cdot)$. Take the first order derivative with respect to θ_i:

$$\frac{\partial H^2(\theta_i)}{\partial \theta_i} = \frac{f(\theta_i)}{F(\theta_i)}\left[N - \sum_{h=1}^{j} k_h - \sum_{h=1}^{j} \frac{k_h F(\theta_i)}{F(\theta_i)+1-F(x_h)} + \frac{F(\theta_i)}{F(\theta_i)+1-F(x_j)}\right]$$

Evaluating the above equation at $\theta_i = x_j$, we obtain:

$$\frac{\partial H^2(\theta_i)}{\partial \theta_i}|_{\theta_i = x_j} = \frac{f(x_j)}{F(x_j)}\left[N - \sum_{h=1}^{j} k_h - \sum_{h=1}^{j} \frac{k_h F(x_j)}{F(x_j)+1-F(x_h)} + F(x_j)\right]$$

Which is greater than zero if and only if:

$$k_j > \frac{1}{1+F(x_j)}\left(N - 1 - \sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)}\right) + 1$$

Because

$$k_j = \left\lceil \frac{1}{1+F(x_j)}\left(N - 1 - \sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)}\right) \right\rceil$$

Therefore

$$k_j \leq \frac{1}{1+F(x_j)}\left(N - 1 - \sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)}\right) + 1$$

and

$$\frac{\partial H^2(\theta_i)}{\partial \theta_i}|_{\theta_i = x_j} \leq 0$$

Furthermore, because

$$N - \sum_{h=1}^{j} k_h - \sum_{h=1}^{j} \frac{k_h F(\theta_i)}{F(\theta_i)+1-F(x_h)} + \frac{F(\theta_i)}{F(\theta_i)+1-F(x_j)} = \frac{\partial H^2(\theta_i)}{\partial \theta_i}\frac{F(\theta_i)}{f(\theta_i)}$$

is negative and decreasing in θ, $\frac{\partial H^2(\theta_i)}{\partial \theta_i} < 0 \; \forall \theta_i < x_j$
Since $H^2(x_j) = 0$ by the definition of x_j, $\forall \theta_i \leq x_j$:

$$H^2(\theta_i) \leq 0 \iff u_i^2(1|\theta_i, \Pi_{-i}^A) \leq u_i^2(2|\theta_i, \Pi_{-i}^A)$$

And the inequality holds strictly if $\theta_i \neq x_j$ or if i is such that $J(i) < j$.

Therefore, it is optimal for any agent i with skill $\theta_i \in [x_{j+1}, x_j)$ to follow $\pi_i^A(\theta_i)$ given that everyone else is following Π_{-i}^A.

Finally, consider an arbitrary agent i with skill $\theta_i < x_{\bar{J}}$. By similar argument to above, it will be optimal for all agents $i \neq N$ to follow $\pi_i(\theta_i) = 0$. Furthermore, for i = N, his payoffs are given by:

$$u_N(1|\theta_N, \Pi_{-i}^A) = w_1 F(\theta_N)^{N-\Sigma_{h=1}^{\bar{J}} k_h - 1} \prod_{h=1}^{\bar{J}} F(x_h)^{k_h} = w_1 \prod_{h=1}^{\bar{J}} F(x_h)^{k_h}$$

$$u_N(2|\theta_N, \Pi_{-i}^A) = w_2 \prod_{h=1}^{\bar{J}} (F(\theta_N)+1-F(x_h))^{k_h}$$

Note that $u_N(1|\theta_N, \Pi_{-i}^A)$ is constant in θ_i while $u_N(2|\theta_N, \Pi_{-i}^A)$ is increasing in θ_N. Furthermore, by the definition of $x_{\bar{J}}$:

$$u_N(1|x_{\bar{J}}, \Pi_{-i}^A) = u_N(2|x_{\bar{J}}, \Pi_{-i}^A)$$

Therefore $\forall \theta_N < x_{\bar{J}}$:

$$u_N(1|\theta_N, \Pi_{-i}^A) > u_N(2|\theta_N, \Pi_{-i}^A)$$

It is therefore optimal for agent N to follow the strategy $\pi_N(\theta_N) = 1$.
Therefore, Π^A is a Nash equilibrium □

Proposition 3:

Proof. We prove this result by focusing on an arbitrary sequence $\mathbb{F}^m$ where $F_i^m(\cdot)$ are all arbitrarily close to some $F(\cdot)$ that is continuously differentiable. We show inductively that for any such $\mathbb{F}^m$, there exists an equilibrium in which:

1. all agents play monotonic pure strategies
2. The threshold at which agents switch from applying to firm 1 to firm 2 converges.

Let $\mathbb{F}^m = \begin{bmatrix} F_1^m(\cdot) \\ \dots \\ F_N^m(\cdot) \end{bmatrix}$ be a sequence of vectors of distributions such that

$$\max_{i \in \{1,\dots,N\}} \{ \sup_{x \in [a,b]} ||F_i^m(x) - F(x)|| \} \to 0$$

$$\max_{i\in\{1,\ldots,N\}}\{\sup_{x\in[a,b]}||f_i^m(x)-f(x)||\}\to 0$$

where $f_i^m(\cdot)$ is the PDF of $F_i^m(\cdot)$, $f(\cdot)$ is the PDF of $F(\cdot)$, and $F_i^m(\cdot)$ are all continuously differentiable for every i and m. For any given $m\in\mathbb{N}$, consider a strategy profile $\Pi^m=\begin{bmatrix}\pi_1^m(\cdot)\\ \ldots\\ \pi_N^m(\cdot)\end{bmatrix}$ where:

$$\pi_i^m(\theta_i)=\begin{cases}1 & \text{if } \theta_i\geq x_i^m\\ 0 & \text{if otherwise}\end{cases}$$

Finally reorder the agents so that:

$$x_i^m\geq x_{i+1}^m\,\forall i\in\{1,\ldots,N-1\}$$

and define x_i^m as: [6]

$$x_i^m=\begin{cases}\max\{x|w_1\prod_{i+1}^N F_h^m(x)=w_2\} & \text{if } i=1\\ a & \text{if } i=N\\ \min\{x\leq x_{i-1}^m|w_1\prod_{i+1}^N F_h^m(\theta)\prod_{h=1}^{i-1}F_h^m(x_h^m)=w_2\prod_{h=1}^{i-1}(F_h^m(\theta)+1-F_h^m(x_h^m))\forall\theta\in[x,x_{i-1}^m]\} & \text{if else}\end{cases}$$

We want to show that for m sufficiently large, Π^m is always an equilibrium of the asymmetric game $\Gamma(N,w_1,w_2,\mathbb{F}^m)$, and that as M goes to infinity, $\Pi^m\to\Pi^A$, where Π^A is the robust equilibrium of the game $\Gamma(N,w_1,w_2,F(\cdot))$

- Base Case: We want to show that:
  1. $x_i^m\to x_1$ if and only if $i\leq k_1$
  2. For m sufficiently large, $\pi_i^m(\theta_i)=\begin{cases}1 & \text{if } \theta_i\geq x_i^m\\ 0 & \text{if } \theta_i<x_i^m\end{cases}$ is optimal for all agents i with $\theta_i\geq x_{k_1+1}^m$

  We once again prove the base case by an inductive argument. We first show that $x_1^m\to x_1$, and that for m sufficiently large,

  $$\pi_1^m(\theta_1)=\begin{cases}1 & \text{if } \theta_1\geq x_1^m\\ 0 & \text{if } \theta_1<x_1^m\end{cases}$$

  is optimal $\forall i$ with $\theta_i\geq x_2^m$
  We then show that assuming the following holds $\forall i<I_1$ where $I_1<k_1$:
  1. $x_i^m\to x_1$ if $i\leq I_1$

[6] Note that by the definition of x_i^m, any agent i with $\theta_i\geq x_i^m$ will always find it optimal to follow the strategy $\pi_i^m(\theta_i)=1$ given Π_{-i}^m. Therefore, to verify that Π^m is in fact a Nash equilibrium, we only need to show that any agent i with $\theta_i<x_i^m$ will also find it optimal to follow the strategy $\pi_i^m(\theta_i)=0$ given Π_{-i}^m

2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for all agents i with $\theta_i \geq x_{I_1+1}^m$

The following is true for $i = I_1 + 1$:

1. $x_i^m \to x_1$
2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for all agents i with $\theta_i \geq x_{I_1+2}^m$

– Base Case:
Claim: $x_1^m \to x_1$:

Proof. Because

$$x_1^m = \inf_{x \in [a,b]} \{x | \pi_1^m(\tilde{x}) = 1 \, \forall \tilde{x} \geq x\}$$

and that $x_1^m \geq \max_{i=2,\ldots,N}\{x_i^m\}$ by assumption, agent 1 with skill $\theta_1 \geq x_2^m$ faces payoffs:

$$u_1(1|\theta_1, \Pi_{-i}^m) = w_1 \prod_{h=2}^{N} F_h^m(\theta_1)$$
$$u_1(2|\theta_1, \Pi_{-i}^m) = w_2$$

In addition, also define the following limiting pseudo-payoff functions:

$$u_1(1|\theta_1, \Pi_{-i}^A) = w_1 F(\theta_1)^{N-1}$$
$$u_1(2|\theta_1, \Pi_{-i}^A) = w_2$$

Further define:

$$H_1^m(\theta_1) = \ln\left(\frac{u_1(1|\theta_1, \Pi_{-i}^m)}{u_1(2|\theta_1, \Pi_{-i}^m)}\right)$$

$$H_1(\theta_1) = \ln\left(\frac{u_1(1|\theta_1, \Pi_{-i}^A)}{u_1(2|\theta_1, \Pi_{-i}^A)}\right)$$

Because:

$$\max_{i \in \{1,\ldots,N\}} \{ \sup_{x \in [a,b]} ||F_i^m(x) - F(x)|| \} \to 0$$

Therefore:

$$\sup_{\theta_1 \in [a,b]} |u_1(1|\theta_1, \Pi_{-i}^m) - u_1(1|\theta_1, \Pi_{-i}^A)| \to 0$$
$$\sup_{\theta_1 \in [a,b]} |u_2(2|\theta_1, \Pi_{-i}^m) - u_2(2|\theta_1, \Pi_{-i}^A)| \to 0$$

Which implies by Triangle Inequality:

$$\sup_{\theta_1 \in [a,b]} |H_1^m(\theta_1) - H_1(\theta_1)| \to 0$$

By definition of x_1:

$$x_1 = \max\{x | H_1(x) = 0\}$$

Similarly,

$$x_1^m = \max\{x | H_1^m(x) = 0\}$$

Now suppose that $x_1^m \not\to x_1$, then at least one of the following must be true:

1. $\exists \delta > 0$ such that there exists a subsequence of $\{x_1^{m_u}\}_{m_u}$ of $\{x_1^m\}_m$ where $x_1^{m_u} > x_1 + \delta \ \forall m_u$
   If this were true, then:

$$\begin{aligned}\sup_{\theta_1 \in [a,b]} |H_1^{m_u}(\theta_1) - H_1(\theta_1)| &\geq |H_1^{m_u}(x_1^{m_u}) - H_1(x_1^{m_u})| \\ &= |H_1(x_1^{m_u})| \\ &\geq \min_{x \geq x_1 + \delta} |H_1(x_1^m)|\end{aligned}$$

   By the Extreme Value Theorem, and the definition of x_1, $\min_{x \geq x_1+\delta} |H_1(x_1^m)|$ is a well defined positive constant. As a result:

$$\sup_{\theta_1 \in [a,b]} |H_1^{m_u}(\theta_1) - H_1(\theta_1)| \not\to 0$$

   Which is a contradiction.
2. $\exists \delta > 0$ such that there exists a subsequence of $\{x_1^{m_d}\}_{m_d}$ of $\{x_1^m\}_m$ where $x_1^{m_d} < x_1 - \delta \ \forall m_d$
   However, note that by the definition of x_1 and the proof of Proposition 2, $\exists \varepsilon < \delta$ such that:

$$H_1(x_1 + \varepsilon) > 0 \text{ and } H_1(x_1 - \varepsilon) < 0$$

   Therefore, $\exists M$ sufficiently large such that $\forall m_d > M$:

$$H_1^{m_d}(x_1 + \varepsilon) > 0 \text{ and } H_1^{m_d}(x_1 - \varepsilon) < 0$$

   Because $H_1^{m_d}(\cdot)$ is a continuous function, there exists $\tilde{x}^{m_d} \in (x_1 - \varepsilon, x_1 + \varepsilon)$ such that $\tilde{x}^m = 0$, which contradicts the definition of x^{m_d}

Therefore, by proof by contradiction:

$$x_1^m \to x_1$$

□

Claim: For m sufficiently large, $\pi_1^m(\theta_1) = 0$ *is optimal for agent 1 given* $\Pi_{-1}^m(\cdot) \ \forall \theta_1 \in [x_2^m, x_1^m)$[7]

[7]Note that this claim is equivalent to stating that $\pi_i^m(\theta_i)$ is optimal for all agents i given $\Pi_{-i}^m \ \forall \theta_i \in [x_2^m, b]$

Proof. This is trivially true as for any such θ_1, agent 1 receives a payoff of

$$w_1 \prod_{h=2}^{N} F_h^m(\theta_1)$$

for applying to firm 1, but a constant payoff of w_2 for applying to firm 2. Furthermore:

$$w_1 \prod_{h=2}^{N} F_h^m(x_1^m) = w_2$$

by the definition of x_1^m. So for any $\theta_1 \in [x_2^m, x_1^m)$:

$$w_1 \prod_{h=2}^{N} F_h^m(\theta_1) < w_2$$

Meaning it is optimal for agent 1 to choose $\pi_1^m(\theta_1) = 0$ □

– Inductive Case:

Proof. Assume the following holds $\forall i < I_1$ where $I_1 < k_1$:

1. $x_i^m \to x_1 \ \forall i \leq I_1$
2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for $\forall i$ given Π_{-i}^m for $\theta_i \in [x_{I_1+1}^m, b]$

Claim: $x_{I_1+1}^m \to x_1$

Proof. Suppose that $x_{I_1+1}^m \not\to x_1$.
Because $x_{I_1+1}^m \leq x_{I_1}^m \ \forall m$ by assumption and that $x_{I_1}^m \to x_1$ by inductive assumption, there must be some $\delta > 0$ such that there exists a subsequence $\{x_{I_1}^{m_u}\}_{m_u}$ such that:

$$x_{I_1+1}^{m_u} < x_1 - \delta \ \forall m_u$$

For $\theta_{I_1+1} \in [x_{I_1+2}^{m_u}, x_{I_1}^{m_u}]$, agent I_1+1 faces payoffs:

$$u_{I_1+1}(1|\theta_{I_1+1}, \Pi_{-\{I_1+1\}}^{m_u}) = w_1 \prod_{h=I_1+2}^{N} F_h^{m_u}(\theta_{I_1+1}) \prod_{h=1}^{I_1} F_h^{m_u}(x_h^{m_u})$$

$$u_{I_1+1}(2|\theta_{I_1+1}, \Pi_{-\{I_1+1\}}^{m_u}) = w_2 \prod_{h=1}^{I_1} (F_h^{m_u}(\theta_{I_1+1}) + 1 - F_h^{m_u}(x_h^{m_u}))$$

With corresponding limiting pseudo-payoffs:

$$u_{I_1+1}(1|\theta_{I_1+1}, \Pi_{-\{I_1+1\}}^{A}) = w_1 F(x_1)^{I_1} \prod_{h=I_1+2}^{N} F(\theta_{I_1+1})$$

$$u_{I_1+1}(2|\theta_{I_1+1}, \Pi_{-\{I_1+1\}}^{A}) = w_2 (F(\theta_{I_1+1}) + 1 - F(x_1))^{I_1}$$

Because $x_i^m \to x_1 \ \forall i \leq I_1$ by inductive assumption:

$$\sup_{\theta_{I_1+1} \in [a,b]} |u_{I_1+1}(1|\theta_{I_1+1}, \Pi^{m_u}_{-\{I_1+1\}}) - u_{I_1+1}(1|\theta_{I_1+1}, \Pi^{A}_{-\{I_1+1\}})| \to 0$$

$$\sup_{\theta_{I_1+1} \in [a,b]} |u_{I_1+1}(2|\theta_{I_1+1}, \Pi^{m_u}_{-\{I_1+1\}}) - u_{I_1+1}(2|\theta_{I_1+1}, \Pi^{A}_{-\{I_1+1\}})| \to 0$$

Note that because $I_1 < k_1$, by the proof of Proposition 2, there exists $\varepsilon \in (0, \delta)$ such that $\forall \theta_{I_1+1} \in [x_1 - \varepsilon, x_1)$:

$$u_{I_1+1}(1|\theta_{I_1+1}, \Pi^{A}_{-\{I_1+1\}}) < u_{I_1+1}(2|\theta_{I_1+1}, \Pi^{A}_{-\{I_1+1\}})$$

Now consider $\tilde{\theta} = x_1 - \frac{\varepsilon}{2}$, and define:

$$\kappa = u_{I_1+1}(2|\theta_{I_1+1}, \Pi^{A}_{-\{I_1+1\}}) - u_{I_1+1}(1|\theta_{I_1+1}, \Pi^{A}_{-\{I_1+1\}})$$

$\exists M$ sufficiently large such that $\forall m_u > M$:

$$|u_{I_1+1}(1|\tilde{\theta}, \Pi^{m_u}_{-\{I_1+1\}}) - u_{I_1+1}(1|\tilde{\theta}, \Pi^{A}_{-\{I_1+1\}})| < \frac{\kappa}{2}$$

$$|u_{I_1+1}(2|\tilde{\theta}, \Pi^{m_u}_{-\{I_1+1\}}) - u_{I_1+1}(2|\tilde{\theta}, \Pi^{A}_{-\{I_1+1\}})| < \frac{\kappa}{2}$$

$$x^{m_u}_{I_1} > \tilde{\theta}$$

As a result, $\forall m_u > M$:

$$u_{I_1+1}(2|\tilde{\theta}, \Pi^{m_u}_{-\{I_1+1\}}) - u^{m_u}_{I_1+1}(1|\tilde{\theta}, \Pi^{m_u}_{-\{I_1+1\}}) > \kappa - \frac{\kappa}{2} - \frac{\kappa}{2} = 0$$

Finally, because $u_{I_1+1}(1|\theta_{I_1+1}, \Pi^{m_u}_{-\{I_1+1\}})$ and $u_{I_1+1}(2|\theta_{I_1+1}, \Pi^{m_u}_{-\{I_1+1\}})$ are both continuous in θ_{I_1+1}, and that by the definition of $x^{m_u}_{I_1}$ and $x^{m_u}_{I_1+1}$:

$$u_{I_1+1}(1|x^{m_u}_{I_1}, \Pi^{m_u}_{-\{I_1+1\}}) - u_{I_1+1}(2|x^{m_u}_{I_1}, \Pi^{m_u}_{-\{I_1+1\}}) > 0$$

Which contradicts the definition of $x^{m_u}_{I_1+1}$
As a result, it cannot be the case that $x^m_{I_1+1} \not\to x_1$, which means that:

$$x^m_{I_1+1} \to x_1$$

□

Claim: For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ *is optimal for* $\forall i$ *given* Π^m_{-i} *for* $\theta_i \in [x^m_{I_1+2}, b]$

Proof. First note that to prove this claim, we only need to show that for agent i with $i \leq I_1 + 1$ and $\theta_i \in [x_{I_1+2}, x_{I_1+1})$, it is optimal for him to follow $\pi_i(\theta_i) = 0$ given Π^m_{-i}
Similar to before, agent i with $\theta_i \in [x_{I_1+2}, x_{I_1+1}]$ faces payoffs:

$$u_i(1|\theta_i, \Pi^m_{-i}) = w_1 \prod_{h=I_1+2}^{N} F^m_h(\theta_i) \prod_{h=1,h\neq i}^{I_1+1} F^m_h(x^m_h)$$
$$u_i(2|\theta_i, \Pi^m_{-i}) = w_2 \prod_{h=1,h\neq i}^{I_1+1} (F^m_h(\theta_i) + 1 - F^m_h(x^m_h))$$

Define:

$$H^m_i(\theta_i) = \ln(\frac{u_i(1|\theta_i, \Pi^m_{-i})}{u_i(2|\theta_i, \Pi^m_{-i})})$$

Taking the first order derivative of $H^m_i(\theta_i)$ with respect to θ_i:

$$\begin{aligned}\frac{\partial H^m_i(\theta_i)}{\partial \theta_i} &= \frac{\partial}{\partial \theta_i} \ln(\prod_{h=I_1+2}^{N} F^m_h(\theta_i)) - \frac{\partial}{\partial \theta_i} \ln(\prod_{h=1,h\neq i}^{I_1+1} (F^m_h(\theta_i) + 1 - F^m_h(x^m_h))) \\ &= \frac{\sum_{l=I_1+2}^{N} f^m_l(\theta_i) \prod_{h=I_1+2,h\neq l}^{N} F^m_h(\theta_i)}{\prod_{h=I_1+2}^{N} F^m_h(\theta_i)} - \frac{\sum_{l=1,l\neq i}^{I_1+1} f^m_l(\theta_i) \prod_{h=1,h\neq i,l}^{I_1+1} (F^m_h(\theta_i) + 1 - F^m_h(x^m_h))}{\prod_{h=1,h\neq i}^{I_1+1} (F^m_h(\theta_i) + 1 - F^m_h(x^m_h))}\end{aligned}$$

Evaluating the equation at $\theta_i = x^m_{I_1+1}$, we obtain:

$$\begin{aligned}\frac{\partial H^m_i(\theta_i)}{\partial \theta_i}|_{\theta_i = x^m_{I_1+1}} &= \frac{\sum_{l=I_1+2}^{N} f^m_l(x^m_{I_1+1}) \prod_{h=I_1+2,h\neq l}^{N} F^m_h(x^m_{I_1+1})}{\prod_{h=I_1+2}^{N} F^m_h(x^m_{I_1+1})} \\ &\quad - \frac{\sum_{l=1,l\neq i}^{I_1+1} f^m_l(x^m_{I_1+1}) \prod_{h=1,h\neq i,l}^{I_1+1} (F^m_h(x^m_{I_1+1}) + 1 - F^m_h(x^m_h))}{\prod_{h=1,h\neq i}^{I_1+1} (F^m_h(x^m_{I_1+1}) + 1 - F^m_h(x^m_h))} \\ &\to \frac{(N - I_1 - 1) f(x_1)}{F(x_1)} - I_1 f(x_1) \\ &= \frac{f(x_1)}{F(x_1)} (N - 1 - I_1(1 + F(x_1))) \\ &> 0\end{aligned}$$

Where the last inequality follows because $I_1 < k_1 = \lceil \frac{N-1}{1+F(x_1)} \rceil \Rightarrow I_1 < \frac{N-1}{1+F(x_1)}$ by assumption.
Furthermore, note that:

$$\sup_{\theta_i \in [a,b]} \{\frac{\partial H^m_i(\theta_i)}{\partial \theta_i} - (\frac{(N - I_1 - 1) f(\theta_i)}{F(\theta_i)} - \frac{I_1 f(\theta_i)}{F(\theta_i) + 1 - F(x_1)})\} \to 0$$

and that $\forall \theta_i < x_1$:

$$\frac{(N - I_1 - 1) f(\theta_i)}{F(\theta_i)} - \frac{I_1 f(\theta_i)}{F(\theta_i) + 1 - F(x_1)} > 0$$

Therefore, $\exists M$ such that $\forall m > M$:

1. $\frac{\partial H_i^m(\theta_i)}{\partial \theta_i}|_{\theta_i = x^m_{I_1+1}} > 0$
2. $\frac{\partial H_i^m(\theta_i)}{\partial \theta_i}|_{\theta_i < x^m_{I_1+1}} > 0$

Because for any $i \leq I_1+1$, $H_i^m(x^m_{I_1+1}) \leq 0$ by definition, $H_i^m(\theta_i) \leq 0 \; \forall \theta_i \in [x^m_{I_1+2}, x^m_{I_1+1}]$. Therefore, it is optimal for agent i with $i \leq I_1+1$ and $\theta_i \in [x_{I_1+2}, x_{I_1+1})$ to follow $\pi_i^m(\theta_i) = 0$ given Π^m_{-i} □

Therefore, the inductive step is complete. □

Therefore, the base case is complete.

- Inductive Case:
We want to show that assuming:

1. $x_i^m \to x_{J(i)} \; \forall i \leq \sum_{h=1}^{j} k_h$
2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for $\forall i$ given Π^m_{-i} for $\theta_i \in [x^m_{\sum_{h=1}^{j} k_h + 1}, b]$

Then the following are true:

1. $x_i^m \to x_{J(i)} \; \forall i \in [\sum_{h=1}^{j} k_h + 1, \sum_{h=1}^{j+1} k_h]$
2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for $\forall i$ given Π^m_{-i} for $\theta_i \in [x^m_{\sum_{h=1}^{j+1} k_h + 1}, b]$

Similar to the base case, we prove the inductive case by an inductive argument. We first show that assuming:

1. $x_i^m \to x_{J(i)} \; \forall i \leq \sum_{h=1}^{j} k_h$
2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for $\forall i$ given Π^m_{-i} for $\theta_i \in [x^m_{\sum_{h=1}^{j} k_h + 1}, b]$

then $x^m_{\sum_{h=1}^{j} k_h + 1} \to x_{j+1}$, and that for m sufficiently large,

$$\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_1 \geq x_i^m \\ 0 & \text{if } \theta_1 < x_i^m \end{cases}$$

is optimal $\forall i$ given Π^m_{-i} for $\theta_i \geq x^m_{\sum_{h=1}^{j} k_h + 2}$

We then show that assuming the following holds $\forall i \in [\sum_{h=1}^{j} k_h + 1, \sum_{h=1}^{j} k_h + I_{j+1}]$ where $I_{j+1} \in [1, k_{j+1})$:

1. $x_i^m \to x_{j+1}$
2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for all agents i with $\theta_i \geq x^m_{\sum_{h=1}^{j} k_h + I_{j+1}+1}$

The following is true for $i = \sum_{h=1}^{j} k_h + I_{j+1} + 1$:

1. $x_i^m \to x_{j+1}$
2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for all agents i with $\theta_i \geq x^m_{\sum_{h=1}^{j} k_h + I_{j+1}+2}$

– Base Case:
Assume:

1. $x_i^m \to x_{J(i)}\ \forall i \leq \sum_{h=1}^{j} k_h$
2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for $\forall i$ given Π^m_{-i} for $\theta_i \in [x^m_{\sum_{h=1}^{j} k_h + 1}, b]$

Define $i^* = \sum_{h=1}^{j} k_h + 1$
Claim: $x_{i^*}^m \to x_{j+1}$:

Proof. Because

$$x_{i^*}^m = \inf_{x \in [a,b]} \{x | \pi_{i^*}^m(\tilde{x}) = 1\ \forall \tilde{x} \geq x\}$$

and that $x_{i^*}^m \geq \max_{i=i^*+1,\ldots,N}\{x_i^m\}$ by assumption, agent i^* with skill $\theta_{i^*} \in [x^m_{i^*+1}, x^m_{i^*-1})$ faces payoffs:

$$u_{i^*}(1|\theta_{i^*}, \Pi^m_{-i^*}) = w_1 \prod_{h=i^*+1}^{N} F_h^m(\theta_{i^*}) \prod_{h=1}^{i^*-1} F_h^m(x_h^m)$$

$$u_{i^*}(2|\theta_{i^*}, \Pi^m_{-i^*}) = w_2 \prod_{h=1}^{i^*-1} [1 - F_h^m(x_h^m) + F_h^m(\theta_i^*)]$$

In addition, also define the following limiting pseudo-payoff functions:

$$u_{i^*}(1|\theta_{i^*}, \Pi^A_{-i^*}) = w_1 F(\theta_{i^*})^{N - \Sigma_{h=1}^{j} k_h - 1} \prod_{h=1}^{j} F(x_h)^{k_h}$$

$$u_{i^*}(2|\theta_{i^*}, \Pi^A_{-i^*}) = w_2 \prod_{h=1}^{j} [1 - F(x_h) + F(\theta_{i^*})]^{k_h}$$

Further define:

$$H_{i^*}^m(\theta_{i^*}) = \ln(\frac{u_{i^*}(1|\theta_{i^*}, \Pi_{-i}^m)}{u_{i^*}(2|\theta_{i^*}, \Pi_{-i}^m)})$$

$$H_{i^*}(\theta_{i^*}) = \ln(\frac{u_{i^*}(1|\theta_{i^*}, \Pi_{-i}^A)}{u_{i^*}(2|\theta_{i^*}, \Pi_{-i}^A)})$$

Because:

$$\max_{i \in \{1,\ldots,N\}} \{ \sup_{x \in [a,b]} ||F_i^m(x) - F(x)|| \} \to 0$$

Therefore:

$$\sup_{\theta_{i^*} \in [a,b]} |u_{i^*}(1|\theta_{i^*}, \Pi_{-i}^m) - u_{i^*}(1|\theta_{i^*}, \Pi_{-i}^A)| \to 0$$
$$\sup_{\theta_{i^*} \in [a,b]} |u_{i^*}(2|\theta_{i^*}, \Pi_{-i}^m) - u_{i^*}(2|\theta_{i^*}, \Pi_{-i}^A)| \to 0$$

Which implies by Triangle Inequality:

$$\sup_{\theta_{i^*} \in [a,b]} |H_{i^*}^m(\theta_{i^*}) - H_{i^*}(\theta_{i^*})| \to 0$$

By definition of x_{j+1}:

$$x_{j+1} = \max\{x | H_{i^*}(x) = 0\}$$

Similarly,

$$x_{i^*}^m = \max\{x | H_{i^*}^m(x) = 0\}$$

Now suppose that $x_{i^*}^m \not\to x_{j+1}$, then at least one of the following must be true:

1. $\exists \delta \in (0, x_j - x_{j+1})$ such that there exists a subsequence of $\{x_{i^*}^{m_u}\}_{m_u}$ of $\{x_{i^*}^m\}_m$ where $x_{i^*}^{m_u} > x_{j+1} + \delta \ \forall m_u$
   If this were true, then:

$$\begin{aligned} \sup_{\theta_{i^*} \in [a,b]} |H_{i^*}^{m_u}(\theta_{i^*}) - H_{i^*}(\theta_{i^*})| &\geq |H_{i^*}^{m_u}(x_{i^*}^{m_u}) - H_{i^*}(x_{i^*}^{m_u})| \\ &= |H_{i^*}(x_{i^*}^{m_u})| \\ &\geq \min_{x \geq x_{j+1} + \delta} |H_{i^*}(x_{i^*}^m)| \end{aligned}$$

   By the Extreme Value Theorem, and the definition of x_{j+1}, $\min_{x \geq x_{j+1}+\delta} |H_{i^*}(x_{i^*}^m)|$ is a well defined positive constant. As a result:

$$\sup_{\theta_{i^*} \in [a,b]} |H_{i^*}^{m_u}(\theta_{i^*}) - H_{i^*}(\theta_{i^*})| \not\to 0$$

   Which is a contradiction.

2. $\exists \delta > 0$ such that there exists a subsequence of $\{x_{i^*}^{m_d}\}_{m_d}$ of $\{x_{i^*}^m\}_m$ where $x_{i^*}^{m_d} < x_{j+1} - \delta \ \forall m_d$
   However, note that by the definition of x_{j+1} and the proof of Proposition 2, $\exists \varepsilon < \delta$ such that:
   $$H_{i^*}(x_{j+1} + \varepsilon) > 0 \text{ and } H_{i^*}(x_{j+1} - \varepsilon) < 0$$
   Therefore, $\exists M$ sufficiently large such that $\forall m_d > M$:
   $$H_{i^*}^{m_d}(x_{j+1} + \varepsilon) > 0 \text{ and } H_{i^*}^{m_d}(x_{j+1} - \varepsilon) < 0$$
   Because $H_{i^*}^{m_d}(\cdot)$ is a continuous function, there exists $\tilde{x}^{m_d} \in (x_{j+1} - \varepsilon, x_{j+1} + \varepsilon)$ such that $\tilde{x}^m = 0$, which contradicts the definition of x^{m_d}
3. There exists a subsequence of $\{x_{i^*}^{m_c}\}_{m_c}$ of $\{x_{i^*}^m\}_m$ where $x_{i^*}^{m_c} > x_j \ \forall m_c$
   If this were true, then by definition of $x_{i^*}^{m_c}$:
   $$x_j < x_{i^*}^{m_c} < x_{\sum_{h=1}^{j} k_h}^{m_c}$$
   by squeeze theorem and the inductive assumption:
   $$x_{i^*}^{m_c} \to x_j$$
   However, consider worker i^* with skill $\theta_{i^*} \in (x_j - \varepsilon, x_j)$ for some small positive ε. For m_c sufficiently large, $\theta_{i^*} < x_{i^*}^m$, and his payoff is given by:
   $$\begin{aligned}
   u_{i^*}(1|\theta_{i^*}, \Pi_{-\{i^*\}}^{m_c}) &= w_1 \prod_{h=i^*+1}^{N} F_h^{m_c}(\max\{x_h^m, \theta_{i^*}\}) \prod_{h=1}^{i^*-1} F_h^{m_c}(x_h^m) \\
   &\geq w_1 \prod_{h=i^*+1}^{N} F_h^{m_c}(\theta_{i^*}) \prod_{h=1}^{i^*-1} F_h^{m_c}(x_h^m) \\
   &\to w_1 F(\theta_{i^*})^{N - \sum_{h=1}^{j} k_h - 1} \prod_{h=1}^{j} F(x_h)^{k_h} \\
   u_{i^*}(2|\theta_{i^*}, \Pi_{-\{i^*\}}^{m}) &= w_2 \prod_{h=1}^{i^*-1} [1 - F_h^m(x_h^m) + F_h^m(\theta_{i^*})] \\
   &\to w_2 \prod_{h=1}^{j} [1 - F(x_h) + F(\theta_{i^*})]^{k_h}
   \end{aligned}$$
   Where the convergence happens by inductive assumption. Now consider
   $$H_{i^*}^{m_c}(\theta) = \ln\left(\frac{u_{i^*}(1|\theta_{i^*}, \Pi_{-\{i^*\}}^{m_c})}{u_{i^*}(2|\theta_{i^*}, \Pi_{-\{i^*\}}^{m_c})}\right)$$
   By the definition of $x_{i^*}^{m_c}$:
   $$H_{i^*}^{m_c}(x_{i^*}^{m_c}) = 0$$

However, by the proof of Proposition 2, we know that there exists some positive ε and δ such that $\forall \theta \in [x_j - \varepsilon, x_j]$:

$$\frac{\partial H_{i^*}^{m_c}(\theta)}{\partial \theta} < -\delta$$

Therefore, for m_c sufficiently large, there exists ε sufficiently small such that the following inequality holds for any $\theta \in (x_{i^*}^m - \varepsilon, x_{i^*}^m)$

$$H_{i^*}^{m_c}(x_{i^*}^{m_c}) > 0 \iff w_1 \prod_{h=i^*+1}^{N} F_h^{m_c}(\theta) \prod_{h=1}^{i^*-1} F_h^{m_c}(x_h^m) > w_2 \prod_{h=1}^{i^*-1} [1 - F_h^m(x_h^m) + F_h^m(\theta)]$$

Which violates the definition of $x_{i^*}^m$

Therefore, by proof by contradiction:

$$x_{i^*}^m \to x_{j+1}$$

□

Claim: For m sufficiently large, $\pi_i^m(\theta_i) = 0$ is optimal for all agents $i < i^*$ given $\Pi_{-i}^m(\cdot)$ $\forall \theta_i \in [x_{i^*+1}^m, x_{i^*}^m)$

Proof. Similar to before, agent i with $\theta_i \in [x_{i^*+1}^m, x_{i^*}^m)$ faces payoffs:

$$u_i(1|\theta_i, \Pi_{-i}^m) = w_1 \prod_{h=i^*+1}^{N} F_h^m(\theta_i) \prod_{h=1, h\neq i}^{i^*} F_h^m(x_h^m)$$

$$u_i(2|\theta_i, \Pi_{-i}^m) = w_2 \prod_{h=1, h\neq i}^{i^*} (F_h^m(\theta_i) + 1 - F_h^m(x_h^m))$$

Define:

$$H_i^m(\theta_i) = \ln\left(\frac{u_i(1|\theta_i, \Pi_{-i}^m)}{u_i(2|\theta_i, \Pi_{-i}^m)}\right)$$

By process similar to the one in the base case, we show that the partial derivative with respect to θ_i evaluated at $\theta_i = x_{i^*}^m$ is:

$$\frac{\partial H_i^m(\theta_i)}{\partial \theta_i}\Big|_{\theta_i = x_{I_1+1}^m} \to f(x_{j+1}) \left[\frac{N - \sum_{h=1}^{j} k_h}{F(x_{j+1})} - \sum_{h=1}^{j} k_h \frac{1}{F(x_{j+1}) + 1 - F(x_h)} - \frac{1}{F(x_{j+1})} \right]$$
$$> 0$$

Furthermore, note that:

$$\sup_{\theta_i \in [a,b]} \left\{ \frac{\partial H_i^m(\theta_i)}{\partial \theta_i} - \frac{f(\theta)}{F(\theta_i)} \left[N - \sum_{h=1}^{j} k_h - 1 - \sum_{h=1}^{j} k_h \frac{F(\theta_i)}{F(\theta_i) + 1 - F(x_h)} \right] \right\} \to 0$$

and that $\forall \theta_i < x_{j+1}$:

$$\left[N - \sum_{h=1}^{j} k_h - 1 - \sum_{h=1}^{j} k_h \frac{F(\theta_i)}{F(\theta_i) + 1 - F(x_h)} \right] > 0$$

Therefore, $\exists M$ such that $\forall m > M$:

1. $\frac{\partial H_i^m(\theta_i)}{\partial \theta_i}|_{\theta_i = x_{i^*}^m} > 0$
2. $\frac{\partial H_i^m(\theta_i)}{\partial \theta_i}|_{\theta_i < x_{i^*}^m} > 0$

Because for any $i < i^*$, $H_i^m(x_{i^*}^m) \leq 0$ by definition, $H_i^m(\theta_i) < 0 \; \forall \theta_i \in [x_{i^*+1}^m, x_{i^*}^m)$. Therefore, it is optimal for agent i with $i \leq i^*$ and $\theta_i \in [x_{i^*+1}^m, x_{i^*}^m)$ to follow $\pi_i^m(\theta_i) = 0$ given Π_{-i}^m □

– Inductive Case:

Proof. Assume the following holds $\forall i \leq \sum_{h=1}^{j} k_h + I_{j+1}$ where $I_{j+1} < k_{j+1}$:

1. $x_i^m \to x_{J(i)}$
2. For m sufficiently large, $\pi_i^m(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x_i^m \\ 0 & \text{if } \theta_i < x_i^m \end{cases}$ is optimal for all agents i with $\theta_i \geq x^m_{\sum_{h=1}^{j} k_h + I_{j+1} + 1}$

Slightly recycling the notation, define $i^* = \sum_{h=1}^{j} k_h + I_{j+1} + 1$ *Claim:* $x_{i^*}^m \to x_{j+1}$

Proof. Suppose that $x_{i^*}^m \not\to x_{j+1}$.
Because $x_{i^*}^m \leq x_{i^*-1}^m \; \forall m$ by assumption and that $x_{i^*-1}^m \to x_{j+1}$ for all $i < i^*$ by inductive assumption, there must be some $\delta > 0$ such that there exists a subsequence $\{x_{I_1}^{m_u}\}_{m_u}$ such that:

$$x_{i^*}^{m_u} < x_{j+1} - \delta \; \forall m_u$$

For $\theta_{i^*} \in [x_{i^*+1}^{m_u}, x_{i^*-1}^{m_u}]$, agent i^* faces payoffs:

$$u_{i^*}(1|\theta_{i^*}, \Pi_{-i^*}^{m_u}) = w_1 \prod_{h=i^*+1}^{N} F_h^{m_u}(\theta_{i^*}) \prod_{h=1}^{i^*-1} F_h^{m_u}(x_h^{m_u})$$

$$u_{i^*}(2|\theta_{i^*}, \Pi_{-i^*}^{m_u}) = w_2 \prod_{h=1}^{i^*-1} (F_h^{m_u}(\theta_{i^*}) + 1 - F_h^{m_u}(x_h^{m_u}))$$

with associated limiting pseudo payoffs:

$$u_{i^*}(1|\theta_{i^*}, \Pi_{-i^*}^{A}) = w_1 F(\theta_{i^*})^{N - \sum_{h=1}^{j} k_h - I_{j+1} - 1} F(x_{j+1})^{I_{j+1}} \prod_{h=1}^{j} F(x_h)^{k_h}$$

$$u_{i^*}(2|\theta_{i^*}, \Pi_{-i^*}^{A}) = w_2 (F(\theta_{i^*}) + 1 - F(x_{j+1}))^{I_{j+1}} \prod_{h=1}^{j} (F(\theta_{i^*}) + 1 - F(x_h))^{k_h}$$

Because $x_i^m \to x_{J(i)} \; \forall i < i^*$ by inductive assumption:

$$\sup_{\theta_{i^*} \in [a,b]} |u_{i^*}(1|\theta_{i^*}, \Pi_{-i^*}^{m_u}) - u_{i^*}(1|\theta_{i^*}, \Pi_{-i^*}^{A})| \to 0$$

$$\sup_{\theta_{i^*} \in [a,b]} |u_{i^*}(2|\theta_{i^*}, \Pi_{-i^*}^{m_u}) - u_{i^*}(2|\theta_{i^*}, \Pi_{-i^*}^{A})| \to 0$$

Note that because $I_{j+1} < k_{j+1}$, by the proof of Proposition 2, there exists $\varepsilon \in (0, \delta)$ such that $\forall \theta_{i^*} \in [x_{j+1} - \varepsilon, x_{j+1})$:

$$u_{i^*}(1|\theta_{i^*}, \Pi^A_{-i^*}) < u_{i^*}(2|\theta_{i^*}, \Pi^A_{-i^*})$$

Now consider $\tilde{\theta} = x_{j+1} - \frac{\varepsilon}{2}$, and define:

$$\kappa = u_{i^*}(2|\theta_{i^*}, \Pi^A_{-i^*}) - u_{i^*}(1|\theta_{i^*}, \Pi^A_{-i^*})$$

$\exists M$ sufficiently large such that $\forall m_u > M$:

$$|u_{i^*}(1|\tilde{\theta}, \Pi^{m_u}_{-i^*}) - u_{i^*}(1|\tilde{\theta}, \Pi^A_{-i^*})| < \frac{\kappa}{2}$$

$$|u_{i^*}(2|\tilde{\theta}, \Pi^{m_u}_{-i^*}) - u_{i^*}(2|\tilde{\theta}, \Pi^A_{-i^*})| < \frac{\kappa}{2}$$

$$x^{m_u}_{I_1} > \tilde{\theta}$$

As a result, $\forall m_u > M$:

$$u_{i^*}(2|\tilde{\theta}, \Pi^{m_u}_{-i^*}) - u^{m_u}_{i^*}(1|\tilde{\theta}, \Pi^{m_u}_{-i^*}) > \kappa - \frac{\kappa}{2} - \frac{\kappa}{2} = 0$$

Finally, because $u_{i^*}(1|\theta_{i^*}, \Pi^{m_u}_{-i^*})$ and $u_{i^*}(2|\theta_{i^*}, \Pi^{m_u}_{-i^*})$ are both continuous in θ_{I_1+1}, and that by the definition of $x^{m_u}_{I_1}$ and $x^{m_u}_{i^*}$:

$$u_{i^*}(1|x^{m_u}_{I_1}, \Pi^{m_u}_{-i^*}) - u_{i^*}(2|x^{m_u}_{I_1}, \Pi^{m_u}_{-i^*}) > 0$$

Which contradicts the definition of $x^{m_u}_{i^*}$
As a result, it cannot be the case that $x^m_{i^*} \not\to x_{j+1}$, which means that:

$$x^m_{i^*} \to x_{j+1}$$

□

Claim: For m sufficiently large, $\pi^m_i(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \geq x^m_i \\ 0 & \text{if } \theta_i < x^m_i \end{cases}$ *is optimal for* $\forall i$ *given* Π^m_{-i} *for* $\theta_i \in [x^m_{i^*+1}, b]$

Proof. Similar to before, agent i with $\theta_i \in [x^m_{i^*+1}, x^m_{i^*})$ faces payoffs:

$$u_i(1|\theta_i, \Pi^m_{-i}) = w_1 \prod_{h=i^*+1}^{N} F^m_h(\theta_i) \prod_{h=1, h\neq i}^{i^*} F^m_h(x^m_h)$$

$$u_i(2|\theta_i, \Pi^m_{-i}) = w_2 \prod_{h=1, h\neq i}^{i^*} (F^m_h(\theta_i) + 1 - F^m_h(x^m_h))$$

Define:

$$H_i^m(\theta_i) = \ln(\frac{u_i(1|\theta_i, \Pi_{-i}^m)}{u_i(2|\theta_i, \Pi_{-i}^m)})$$

By process similar to the one in the base case, we show that the partial derivative with respect to θ_i evaluated at $\theta_i = x_{i^*}^m$ is:

$$\frac{\partial H_i^m(\theta_i)}{\partial \theta_i}|_{\theta_i = x_{I_1+1}^m} \to f(x_{j+1}) \left[\frac{N - \sum_{h=1}^{j} k_h}{F(x_{j+1})} + 1 - \sum_{h=1}^{j} k_h \frac{1}{F(x_{j+1}) + 1 - F(x_h)} - I_{j+1} \frac{1 + F(x_{j+1})}{F(x_{j+1})} \right]$$
$$> 0$$

Furthermore, note that:

$$\sup_{\theta_i \in [a,b]} \{ \frac{\partial H_i^m(\theta_i)}{\partial \theta_i} - \frac{f(\theta)}{F(\theta_i)} \left[N - \sum_{h=1}^{j} k_h - I_{j+1} - \sum_{h=1}^{j} \frac{k_h F(\theta_i)}{F(\theta_i) + 1 - F(x_h)} - \frac{F(\theta_i)(I_{j+1} - 1)}{F(\theta_i) + 1 - F(x_{j+1})} \right] \} \to 0$$

and that $\forall \theta_i < x_{j+1}$:

$$N - \sum_{h=1}^{j} k_h - I_{j+1} - \sum_{h=1}^{j} k_h \frac{F(\theta_i)}{F(\theta_i) + 1 - F(x_h)} - \frac{F(\theta_i)(I_{j+1} - 1)}{F(\theta_i) + 1 - F(x_{j+1})} > 0$$

Therefore, $\exists M$ such that $\forall m > M$:

1. $\frac{\partial H_i^m(\theta_i)}{\partial \theta_i}|_{\theta_i = x_{i^*}^m} > 0$
2. $\frac{\partial H_i^m(\theta_i)}{\partial \theta_i}|_{\theta_i < x_{i^*}^m} > 0$

Because for any $i < i^*$, $H_i^m(x_{i^*}^m) \leq 0$ by definition, $H_i^m(\theta_i) < 0 \; \forall \theta_i \in [x_{i^*+1}^m, x_{i^*}^m)$.
Therefore, it is optimal for agent i with $i \leq i^*$ and $\theta_i \in [x_{i^*+1}^m, x_{i^*}^m)$ to follow $\pi_i^m(\theta_i) = 0$ given Π_{-i}^m □

Therefore, the inductive step is complete. □

Therefore, the inductive step is complete.

And by a nested proof by induction, we have shown that Π^A is a robust equilibrium □

Proposition 4:

Proof. We show inductively that the conditions for a Nash equilibrium uniquely pin down the strategy profiles of individual agents (up to some reorderings). First note that given any two pure monotonic strategy profiles $\tilde{\Pi}$ and $\hat{\Pi}$ with associated sets of group sizes and thresholds $\{(\tilde{x}_j, \tilde{k}_j)\}$ and $\{(\hat{x}_j, \hat{k}_j)\}$, $\tilde{\Pi}$ and $\hat{\Pi}$ are reorderings of each other if and only if $\{(\tilde{x}_j, \tilde{k}_j)\}$ and $\{(\hat{x}_j, \hat{k}_j)\}$ are identical. As a result, to prove Proposition 4, we only need to show that all pure strategy equilibria are monotonic and that equilibrium conditions uniquely determine the set of group sizes and thresholds.

Formally: let Π be a pure strategy Nash equilibrium of the game $\Gamma(N, w_1, w_2, F(\cdot))$, and define $\theta_i^* = \inf_{\theta \in [a,b]} \{\theta | \pi_i(x) = 1 \forall x \geq \theta\}$. Let the set $\{x_j\}_{j=1}^{\bar{J}}$ be the set of unique elements of the

set $\{\theta_i^*\}_{i=1,...,N}$, and assume W.L.O.G. that $x_j > x_{j+1}$. Let $I_j = \{i \in \{1,...,N\} | \theta_i^* = x_j\}$, and let $k_j = ||I_j||$ be the size of set I_j. We want to show that (x_j, k_j) are identical for any pure strategy Nash equilibrium Π, and that $\pi_i(\theta_i) = 0 \; \forall \theta_i < \theta_i^* \forall i$. We do so by the following inductive argument:

- Base Case: First note that in any equilibrium, agents with sufficiently high θ_i will always find it optimal to follow $\pi_i(\theta_i) = 1$. Therefore, there exists some x_1 such that $\pi_i(\theta_i) = 1$ is optimal for all agents i with $\theta_i \geq x_1$
  Consider an arbitrary agent i with skill $\theta_i \geq x_1$. His payoffs from applying to firm 1 and 2 are given respectively by:

$$u_i(1|\theta_i, \Pi_{-i}) = w_1 F(\theta_i)^{N-1}$$
$$u_i(2|\theta_i, \Pi_{-i}) = w_2$$

  x_1 is therefore determined by the condition:

$$w_1 F(x_1)^{N-1} = w_2$$

  Which uniquely determines x_1. In other words, in any pure strategy Nash equilibrium Π, $\pi_i(\theta_i) = 1 \; \forall i \; \forall \theta_i \geq x_1$.
  Now consider an agent with $\theta_i \in [\max x_2, a, x_1]$.

  - If $i \in I_1$, then agent i faces payoffs:

$$u_i^2(1|\theta_i, \Pi_{-i}) = w_1 F(\theta_i)^{N-k_1} F(x_1)^{k_1-1}$$
$$u_i^2(2|\theta_i, \Pi_{-i}) = w_2 (F(\theta_i) + 1 - F(x_1))^{k_1-1}$$

    Individual rationality requires that:

$$u_i^2(1|\theta_i, \Pi_{-i}) \leq u_i^2(2|\theta_i, \Pi_{-i}) \; \forall \theta_i < x_1$$

    Which requires that:

$$\frac{\partial}{\partial \theta_i}[u_i^2(1|\theta_i, \Pi_{-i}) - u_i^2(2|\theta_i, \Pi_{-i})]|_{\theta_i = x_1} > 0$$

    With a bit of algebra (see the proof of Proposition 2), it can be shown that the above condition can be simplified to:

$$k_1 < \frac{N-1}{1 + F(x_1)} + 1$$

  - If $i \notin I_1$, then agent i faces payoffs:

$$u_i^1(1|\theta_i, \Pi) = w_1 F(\theta_i)^{N-k_1-1} F(x_1)^{k_1}$$
$$u_i^1(2|\theta_i, \Pi) = w_2 (F(\theta_i) + 1 - F(x_1))^{k_1}$$

    Individual rationality requires that:

$$u_i^1(1|\theta_i, \Pi_{-i}) \geq u_i^1(2|\theta_i, \Pi_{-i}) \; \forall \theta_i < x_1$$

Which requires that:

$$\frac{\partial}{\partial \theta_i}[u_i^1(1|\theta_i,\Pi_{-i}) - u_i^1(2|\theta_i,\Pi_{-i})]|_{\theta_i=x_1} < 0$$

With a bit of algebra, it can be shown that the above condition can be simplified to:

$$k_1 > \frac{N-1}{1+F(x_1)}$$

Therefore, $k_1 \in (\frac{N-1}{1+F(x_1)}, \frac{N-1}{1+F(x_1)} + 1)$, meaning that there exists one and only one k_1 such that individual rationality could be satisfied for all agents with $\theta_i \in (x_2, x_1)$.
Furthermore, note that by the proof of Proposition 2, k_1 is such that:

- For agent $i \in I_1$:
$$u_i^2(1|\theta_i,\Pi_{-i}) < u_i^2(2|\theta_i,\Pi_{-i}) \ \forall \theta_i < x_1$$
Therefore, in any pure strategy Nash equilibrium Π, any agent $i \in I_1$ strictly prefers following $\pi_i(\theta_i) = 0 \ \forall \theta_i \in (x_2, x_1)$
- For agent $i \notin I_1$:
$$u_i^1(1|\theta_i,\Pi_{-i}) > u_i^1(2|\theta_i,\Pi_{-i}) \ \forall \theta_i \in (x_2, x_1)$$
Therefore, in any pure strategy Nash equilibrium Π, any agent $i \notin I_1$ strictly prefers following $\pi_i(\theta_i) = 1 \ \forall \theta_i \in (x_2, x_1)$

Therefore, (x_1, k_1) is identical in any pure strategy Nash equilibrium and $\pi_i(\theta_i)$ is weakly increasing for all i on the interval $(x_2, b]$

- Inductive Case:
Assume that (x_h, k_h) is identical for all pure strategy Nash equilibria $\forall h < j$ and that $\pi_i(\theta_i)$ is weakly increasing for all $\theta_i \in (x_j, b) \ \forall i$

By the definition of x_j, x_j must be the largest solution of:

$$w_1 F(x_j)^{N-1-\Sigma_{h=1}^{j-1} k_h} \prod_{h=1}^{j-1} F(x_h)^{k_h} = w_2 \prod_{h=1}^{j-1} (F(x_j) + 1 - F(x_h))^{k_h}$$

That is still smaller than x_{j-1}. The above equation states that x_j must be such that any worker i with $\pi_i(\theta_i) = 1 \forall \theta_i \geq x_j$ is indifferent between applying to firm 1 and firm 2 if $\theta_i = x_j$ (the LHS is the payoff from applying to firm 1, and the RHS is the payoff from applying to firm 2). Note that by the definition of x_{j-1} and x_j, the LHS is necessarily greater than the RHS at $x_j = x_{j-1}$, and the LHS is smaller than the RHS at $x_j = a$. As a result, by the intermediate value theorem, a solution for x_j is guaranteed to exist on the interval $[a, x_{j-1}]$.

Note that given $\{(x_h, k_h)\}_{h=1,\ldots,j-1}$, the above equation is deterministic. As a result, x_j is always uniquely defined. Now consider an agent i with $\theta_i < (x_{j+1}, x_j)$.

– If $i \in I_h$ for some $h \leq j$, then agent i faces payoffs:

$$u_i^2(1|\theta_i, \Pi_{-i}) = w_1 F(\theta_i)^{N-\Sigma_{h=1}^{j} k_h} \frac{1}{F(x_t)} \prod_{h=1}^{j} F(x_h)^{k_h}$$

$$u_i^2(2|\theta_i, \Pi_{-i}) = w_2 \frac{1}{F(\theta_i)+1-F(x_t)} \prod_{h=1}^{j} (F(\theta_i)+1-F(x_h))^{k_h}$$

Individual rationality requires that:

$$u_i^2(1|\theta_i, \Pi_{-i}) \leq u_i^2(2|\theta_i, \Pi_{-i}) \; \forall \theta_i < x_1$$

Which requires that:

$$\frac{\partial}{\partial \theta_i}[u_i^2(1|\theta_i, \Pi_{-i}) - u_i^2(2|\theta_i, \Pi_{-i})]|_{\theta_i = x_j} > 0$$

With a bit of algebra (see the proof of Proposition 2), it can be shown that the above condition can be simplified to:

$$k_j < \frac{1}{1+F(x_j)}(N-1-\sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)}) + 1$$

– If $i \notin \cup_{h=1}^{j} I_h$, then agent i faces payoffs:

$$u_i^1(1|\theta_i, \Pi_{-i}) = w_1 F(\theta_i)^{N-1-\Sigma_{h=1}^{j} k_h} \prod_{h=1}^{j} F(x_h)^{k_h}$$

$$u_i^1(2|\theta_i, \Pi_{-i}) = w_2 \prod_{h=1}^{j} (F(\theta_i)+1-F(x_h))^{k_h}$$

Individual rationality requires that:

$$u_i^1(1|\theta_i, \Pi_{-i}) \geq u_i^1(2|\theta_i, \Pi_{-i}) \; \forall \theta_i < x_1$$

Which requires that:

$$\frac{\partial}{\partial \theta_i}[u_i^1(1|\theta_i, \Pi_{-i}) - u_i^1(2|\theta_i, \Pi_{-i})]|_{\theta_i = x_j} < 0$$

With a bit of algebra, it can be shown that the above condition can be simplified to:

$$k_j > \frac{1}{1+F(x_j)}(N-1-\sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)})$$

Therefore, $k_j \in (\frac{1}{1+F(x_j)}(N-1-\sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)}), \frac{1}{1+F(x_j)}(N-1-\sum_{h=1}^{j-1} k_h \frac{2F(x_j)+1-F(x_h)}{F(x_j)+1-F(x_h)}) + 1)$, meaning that there exists one and only one k_j such that individual rationality could be satisfied for all agents with $\theta_i < x_j$.
Furthermore, note that by the proof of Proposition 2, k_j is such that:

– For agent $i \in I_h$ for some $h \leq j$:

$$u_i^2(1|\theta_i, \Pi_{-i}) < u_i^2(2|\theta_i, \Pi_{-i}) \; \forall \theta_i < x_j$$

Therefore, in any pure strategy Nash equilibrium Π, any agent $i \in \cup_{h=1}^{j} I_h$ strictly prefers following $\pi_i(\theta_i) = 0 \; \forall \theta_i \in (x_{j+1}, x_j)$

– For agent $i \notin \cup_{h=1}^{j} I_h$:

$$u_i^1(1|\theta_i, \Pi_{-i}) > u_i^1(2|\theta_i, \Pi_{-i}) \; \forall \theta_i \in (x_{j+1}, x_j)$$

Therefore, in any pure strategy Nash equilibrium Π, any agent $i \notin \cup_{h=1}^{j} I_h$ strictly prefers following $\pi_i(\theta_i) = 1 \; \forall \theta_i \in (x_{j+1}, x_j)$

Assume that (x_h, k_h) is identical for all pure strategy Nash equilibria $\forall h \leq j$, and $\pi_i(\theta_i)$ is weakly increasing for all $\theta_i \in (x_{j+1}, b) \; \forall i$

By the inductive argument, (x_j, k_j) are identical for any pure strategy Nash equilibrium Π, and that $\pi_i(\theta_i)$ is weakly increasing $\forall i$. As a result, all pure strategy Nash equilibria are monotonic and equivalent under reordering. □

Proposition 5:

Proof. The proof to this proposition is simply an inductive argument showing that for m sufficiently large, optimality conditions recursively pin down x_i^m and require that agent i with $\theta_i < x_i^m$ finds it strictly optimal to choose $\pi_i^m(\theta_i) = 0$. This proof refers heavily to the one for Proposition 3.

- Base Case:
  Similar to the proof of Proposition 3, in any equilibrium, x_1^m is uniquely defined by:

$$x_1^m = \max_{x \in [a,b]} \{x | w_1 \prod_{h=2}^{N} F_h(x) = w_2\}$$

And as shown in the proof for Proposition 3, $x_1^m \to x_1$ as $m \to \infty$.
Now consider agent 1 with $\theta_1 \in [x_2^m, x_1^m]$, his payoffs are:

$$u_1^m(1|\theta_i, \Pi_{-1}^m) = w_1 \prod_{h=2}^{N} F_h(\theta_i)$$
$$u_1^m(2|\theta_i, \Pi_{-1}^m) = w_2$$

Therefore, in any Nash equilibrium, optimality requires that $\pi_1(\theta_1) = 0 \; \forall \theta_i \in [x_2, 1)$

- Inductive Case:
  Assume that for all $i \leq I$:

  – x_i^m is uniquely defined by optimality conditions and $x_i^m \to x_{J(i)}$, where $J(i)$ is defined in Proposition 2

– For m sufficiently large, $\pi_i^m(\theta_i) = 0 \forall \theta_i \in [x_{I+1}^m, x_i^m)$ in any Nash equilibrium

First note that by definition of x_{I+1}^m, for m sufficiently large, x_{I+1}^m is defined by the following condition:

$$x_{I+1}^m = \max\{x | w_1 \prod_{h=1}^{I_1} F_h^m(x_h^m) \prod_{h=I_1+2}^{N} F_h^m(x) = w_2 \prod_{h=1} (F_h^m(x) + 1 - F_h^m(x))\}$$

The above condition uniquely determines x_{I+1}^m given $\{x_1^m, ..., x_I^m\}$, which are also uniquely determined in equilibrium by inductive assumptions. Furthermore, by argument similar to the one in the proof for Proposition 3, $x_{I+1}^m \to x_{J(I+1)}$
Now consider agent i with $i \geq I+1$ and $\theta_i \in [x_{I+2}^m, x_{I+1}^m)$, agent i faces the following payoffs:

$$u_1^m(1|\theta_i, \Pi_{-1}^m) = w_1 \prod_{h=1}^{I} F_h^m(x_h^m) \prod_{h=I+2}^{N} F_h^m(x)$$

$$u_1^m(2|\theta_i, \Pi_{-1}^m) = w_2 \prod_{h=1}^{I} (F_h^m(x) + 1 - F_h^m(x))$$

By argument similar to the one used in the proof of Proposition 3:

$$u_1^m(1|\theta_i, \Pi_{-1}^m) < u_1^m(2|\theta_i, \Pi_{-1}^m) \ \ \forall \theta_i \in [x_{I+2}^m, x_{I+1}^m)$$

. Therefore, in any Nash equilibrium, it is strictly optimal for agent i with $i \leq I+1$ and $\theta_i \in [x_{I+2}, x_{I+1})$ to follow $\pi_i(\theta_i) = 0$ given Π_{-i}^m

□

Lemma 1:

Proof. Consider a sequence of distributions $\{F^m(\cdot)\}_{k=1}^{\infty}$:

$$F^m(\cdot) = \begin{bmatrix} F_1^m(\cdot) \\ ... \\ F_N^m(\cdot) \end{bmatrix}$$

Such that:

1. $\max_{i \in \{1,...,N\}} \{\sup_{x \in [a,b]} ||F_i^m(x) - F(x)||\} \to 0$ as $m \to \infty$
2. $\max_{i \in \{1,...,N\}} \{\sup_{x \in [a,b]} ||f_i^m(x) - f(x)||\} \to 0$ as $m \to \infty$[8]
3. $F_i(\cdot)$ FOSD $F_{i-1}(\cdot)$ $\forall i \in \{2, ..., N\}$

Let Π^m be a Nash equilibrium, and define

$$x_i^m = \inf\{x | \pi_i^m(\theta_i) = 1 \ \ \forall \theta_i \geq x\}$$

We show inductively that $x_1^m > x_2^m > ... > x_N^m$.

[8] where $f_i^m(\cdot)$ is the PDF of $F_i^m(\cdot)$, $f(\cdot)$ is the PDF of $F(\cdot)$, and $F_i^m(\cdot)$ are all continuously differentiable for every i and m.

- Base Case:
  Let $\tilde{x}_1^m = \max_{x_1^m,\ldots,x_N^m}$ and suppose that $x_1^m \neq \tilde{x}_1^m$.
  Assume W.L.O.G. that $\tilde{x}_1^m = x_i^m$ where $i \neq 1$. Note that x_i^m in this case solves the following equation:

$$w_1 \prod_{h=1,h\neq i}^{N} F_h^m(x_i^m) = w_2$$

  Now consider agent 1 with $\theta_1 = x_i^m$. By assumption, $\theta_1 > x_1^m$. However, at $\theta_1 = x_i^m$, agent 1 faces payoffs:

$$\begin{aligned} u_1^m(1|x_i^m,\Pi^m) &= w_1 \prod_{h=2}^{N} F_h^m(x_i^m) \\ &= w_1 \frac{F_i^m(x_i^m)}{F_1^m(x_i^m)} \prod_{h=1,h\neq i}^{N} F_h^m(x_i^m) \\ &< w_1 \prod_{h=1,h\neq i}^{N} F_h^m(x_i^m) \\ &= w_2 = u_1^m(2|x_i^m,\Pi^m) \end{aligned}$$

  Where the inequality follows by FOSD assumption. Therefore, individual optimality requires $\pi_1^m(x_i^m) = 0$, which violates the definition of x_i^m.
  Therefore, $\tilde{x}_1^m = \max_{x_1^m,\ldots,x_N^m} = x_1^m$.
  Furthermore, note that if $\exists i \in \{2,\ldots,N\}$ such that $x_i^m = x_1^m$, then x_i^m must be such that:

$$w_1 \prod_{h=1,h\neq i}^{N} F_h^m(x_i^m) = w_1 \prod_{h=1,h\neq i}^{N} F_h^m(x_1^m) = w_2$$

  Which is impossible because in equilibrium:

$$w_1 \prod_{h=2}^{N} F_h^m(x_1^m) = w_2$$

  and $F_1^m(x_1^m) \neq F_i^m(x_1^m)\ \forall i \neq 1$ by FOSD assumption

- Inductive Case:
  Suppose $x_1^m > x_2^m > \ldots > x_{I-1}^m$. Let $\tilde{x}_I^m = \max_{x_I^m,\ldots,x_N^m}$ and suppose that $x_I^m \neq \tilde{x}_I^m$.
  Assume W.L.O.G. that $\tilde{x}_I^m = x_i^m$ where $i > I$. Note that x_i^m in this case is the largest solution to the following equation:

$$\begin{aligned} w_1 \prod_{h=I,h\neq i}^{N} F_h^m(x_i^m) \prod_{h=1}^{I-1} (F_h^m(x_i^m) + \int_{x_i^m}^{x_h^m} (1-\pi_h^m(\theta)) f_h^m(\theta) d\theta) \\ = w_2 \prod_{h=1}^{I-1} (F_h^m(x_i^m) + 1 - F_h^m(x_h^m) + \int_{x_i^m}^{x_h^m} \pi_h^m(\theta) f_h^m(\theta) d\theta) \end{aligned}$$

Now consider agent I with $\theta_I = x_i^m$. By assumption, $\theta_I > x_I^m$. However, at $\theta_I = x_i^m$, agent I faces payoffs:

$$\begin{aligned}
u_I^m(1|x_i^m,\Pi^m) &= w_1 \prod_{h=I+1}^{N} F_h^m(x_i^m) \prod_{h=1}^{I-1}(F_h^m(x_i^m) + \int_{x_i^m}^{x_h^m}(1-\pi_h^m(\theta))f_h^m(\theta)d\theta) \\
&= \frac{F_i^m(x_i^m)}{F_I^m(x_i^m)} w_1 \prod_{h=I,h\neq i}^{N} F_h^m(x_i^m) \prod_{h=1}^{I-1}(F_h^m(x_i^m) + \int_{x_i^m}^{x_h^m}(1-\pi_h^m(\theta))f_h^m(\theta)d\theta) \\
&= \frac{F_i^m(x_i^m)}{F_I^m(x_i^m)} w_2 \prod_{h=1}^{I-1}(F_h^m(x_i^m) + 1 - F_h^m(x_h^m) + \int_{x_i^m}^{x_h^m}\pi_h^m(\theta)f_h^m(\theta)d\theta) \\
&< w_2 \prod_{h=1}^{I-1}(F_h^m(x_i^m) + 1 - F_h^m(x_h^m) + \int_{x_i^m}^{x_h^m}\pi_h^m(\theta)f_h^m(\theta)d\theta) = u_I^m(2|x_i^m,\Pi^m)
\end{aligned}$$

Where the inequality follows by FOSD assumption. Therefore, individual optimality requires $\pi_I^m(x_i^m) = 0$, which violates the definition of x_I^m.
Therefore, $\tilde{x}_I^m = \max_{x_I^m,...,x_N^m} = x_I^m$.
Furthermore, note that if $\exists i \in \{I+1,...,N\}$ such that $x_i^m = x_I^m$, then x_i^m must be such that:

$$\begin{aligned}
u_i^m(1|x_i^m,\Pi^m) &= w_1 \prod_{h=I,h\neq i}^{N} F_h^m(x_i^m) \prod_{h=1}^{I-1}(F_h^m(x_i^m) + \int_{x_i^m}^{x_h^m}(1-\pi_h^m(\theta))f_h^m(\theta)d\theta) \\
&= w_2 \prod_{h=1}^{I-1}(F_h^m(x_i^m) + 1 - F_h^m(x_h^m) + \int_{x_i^m}^{x_h^m}\pi_h^m(\theta)f_h^m(\theta)d\theta) \\
&= w_1 \prod_{h=I+1}^{N} F_h^m(x_i^m) \prod_{h=1}^{I-1}(F_h^m(x_i^m) + \int_{x_i^m}^{x_h^m}(1-\pi_h^m(\theta))f_h^m(\theta)d\theta)
\end{aligned}$$

Which is again impossible in equilibrium since $F_I^m(x_I^m) \neq F_i^m(x_I^m)\ \forall i \neq I$ by FOSD assumption

Therefore, if $\{F^m(\cdot)\}_{k=1}^{\infty}$ is a sequence of vectors of distributions such that:

1. $\max_{i\in\{1,...,N\}}\{\sup_{x\in[a,b]}||F_i^m(x) - F(x)||\} \to 0$ as $m \to \infty$
2. $\max_{i\in\{1,...,N\}}\{\sup_{x\in[a,b]}||f_i^m(x) - f(x)||\} \to 0$ as $m \to \infty$[9]
3. $F_i(\cdot)$ FOSD $F_{i-1}(\cdot)\ \forall i \in \{2,...,N\}$

then $x_1^m > x_2^m > ... > x_N^m$ in every Nash equilibrium Π^m, which is the claim of Lemma 1. □

Proposition 7:

Proof. We first note that the proofs for Propositions 2 through 5 apply to any function $F(\cdot)$ that is continuous, positive, and weakly increasing. As a result, if we can show that in any monotonic

[9]where $f_i^m(\cdot)$ is the PDF of $F_i^m(\cdot)$, $f(\cdot)$ is the PDF of $F(\cdot)$, and $F_i^m(\cdot)$ are all continuously differentiable for every i and m.

pure strategy profile Π of the replicator economy, the payoff of workers can be written in the form of:

$$u_i(a_i|\theta_i,\Pi_{-i}) = \begin{cases} \kappa \frac{1}{G\left(\max\{\theta_i, x_{J^R(i)}\}\right)} \prod_{h=1}^{j-1} G(x_h)^{k_h} \prod_{h=j}^{\bar{J}} G(\theta_i)^{k_h} & \text{if } a_i = 1 \\ w_2 \frac{1}{1-G(x_{J(i)})+G(\min\{\theta_i, x_{J^R(i)}\})} \prod_{h=1}^{j-1} (1-G(x_h)+G(\theta_i))^{k_h} & \text{if } a_i = 2 \end{cases}$$

for some positive monotonic continuous function $G(\cdot)$ and positive constant κ, then we can directly apply the proofs of Propositions 2 through 5 to the replicator economy with minimal modification. This allows us to show that all robust equilibria are necessarily monotonic pure strategy equilibria and that all pure strategy equilibria are reorderings of each other. For the sake of brevity, we do not present a complete recreation of the proofs of Propositions 2 through 6 (and Lemma 1).

Let Π be a monotonic pure strategy profile, and let $\{(x_j,k_j)\}_{j=1,\dots,\bar{J}}$ be the thresholds and sizes of each group. Assume W.L.O.G. that $x_j > x_{j+1} \forall j$. Consider a worker i with skill $\theta_i \in [x_j, x_{j-1}]$, and define $x_0 = b$. Worker i's equilibrium payoffs are given by:

$$\begin{aligned} u_i(1|\theta_i,\Pi_{-i}) &= \frac{1}{1-\frac{1-F(\max\{\theta_i,x_{J^R(i)}\})}{H_1}} \left[\prod_{h=1}^{j-1}(1-\frac{1-F(x_h)}{H_1})^{k_h} \prod_{h=j}^{\bar{J}}(1-\frac{1-F(\theta_i)}{H_1})^{k_h}\right] \\ &= \frac{1}{\frac{H_1-1}{H_1}+\frac{F(\max\{\theta_i,x_{J^R(i)}\})}{H_1}} \left[\prod_{h=1}^{j-1}(\frac{H_1-1}{H_1}+\frac{F(x_h)}{H_1})^{k_h} \prod_{h=j}^{\bar{J}}(\frac{H_1-1}{H_1}+\frac{F(\theta_i)}{H_1})^{k_h}\right] \\ &= \left(\frac{H_2}{H_1}\right)^{N-1} \frac{1}{\frac{H_1-1}{H_2}+\frac{F(\max\{\theta_i,x_{J^R(i)}\})}{H_2}} \left[\prod_{h=1}^{j-1}(\frac{H_1-1}{H_2}+\frac{F(x_h)}{H_2})^{k_h} \prod_{h=j}^{\bar{J}}(\frac{H_1-1}{H_1}+\frac{F(\theta_i)}{H_2})^{k_h}\right] \\ &= \kappa \frac{1}{G\left(\max\{\theta_i,x_{J^R(i)}\}\right)} \left[\prod_{h=1}^{j-1} G(x_h)^{k_h} \prod_{h=j}^{\bar{J}} G(\theta_i)^{k_h}\right] \\ u_i(2|\theta_i,\Pi_{-i}) &= w_2 \frac{1}{1-\frac{F(x_{J(i)})-F(\min\{\theta_i,x_{J(i)}\})}{H_2}} \left[\prod_{h=1}^{j-1}(1-\frac{F(x_h)-F(\theta_i)}{H_2})^{k_h}\right] \\ &= w_2 \frac{1}{1-\frac{F(x_{J(i)})}{H_2}+\frac{F(\min\{\theta_i,x_{J^R(i)}\})}{H_2}} \left[\prod_{h=1}^{j-1}(1-\frac{F(x_h)}{H_2}+\frac{F(\theta_i)}{H_2})^{k_h}\right] \\ &= w_2 \frac{1}{1-G(x_{J(i)})+G(\min\{\theta_i,x_{J^R(i)}\})} \left[\prod_{h=1}^{j-1}(1-G(x_h)+G(\theta_i))^{k_h}\right] \end{aligned}$$

Where $G(\theta) = \frac{H_1-1}{H_2} + \frac{F(\theta)}{H_2}$ and $\kappa = \left(\frac{H_2}{H_1}\right)^{N-1}$. It is obvious that $G(\theta)$ is a continuous monotonically increasing function that is always positive. As a result, all the proofs for Propositions 2 through 6 can be applied to the replicator economy. □

Proposition 8:

Proof. It is easy to check that if $w_2 \leq e^{-\frac{1}{\alpha_1}}$, then $x_1^N \to 0$, and $\zeta^N(\theta) = \zeta^*(\theta) = 1$ holds trivially in the limit. Therefore, we assume that $w_2 > e^{-\frac{1}{\alpha_1}}$ so that $x_1^N > 0$ for sufficiently large N. First, it

can be seen that for any $N \in \mathbb{N}$, $\zeta^N(\theta) = \zeta^*(\theta)\ \forall \theta > \max\{x_1^N, x_1\}$. Furthermore, by equilibrium definition, x_1^N solves:

$$w_2 = (1 - \frac{1 - F(x_1^N)}{\alpha_1 N})^{N-1}$$

Taking the limit on both sides, we obtain the limit of x_1^N as:

$$x_1^N \to x_1 = F^{-1}\left(1 + \alpha_1 \ln(w_2)\right)$$

Furthermore, for $\theta \in [x_2^N, x_1^N]$:

$$\zeta^N(\theta) = \frac{N - k_1}{N} = 1 - \frac{1}{N}\lceil \frac{N-1}{1 + \frac{\alpha_1}{\alpha_2} w_2^{\frac{1}{N-1}}} \rceil \to \frac{\alpha_1}{\alpha_1 + \alpha_2} = \zeta^*(\theta)$$

Now suppose $\exists \tilde{\theta} < x_1$ such that $\zeta^N(\tilde{\theta}) \not\to \zeta^*(\tilde{\theta})$. By definition of $\zeta^N()$, $\lim_{N\to\infty} \zeta^N(\theta) \leq \zeta^*(\theta)\ \forall \theta \in [a,b]$. Therefore, $\zeta^N(\tilde{\theta}) \not\to \zeta^*(\tilde{\theta})$ implies that $\exists \delta > 0$ such that $\zeta^N(\tilde{\theta}) < \zeta^*(\tilde{\theta}) - \delta$ for infinitely many $N \in \mathbb{N}$.

Therefore, consider the sequence $\{\zeta^M(\tilde{\theta})\}$ that only contains elements of $\{\zeta^N(\tilde{\theta})\}$ that are smaller or equal to $\zeta^*(\tilde{\theta}) - \delta$. Because $\zeta^M(\tilde{\theta}) \in [0, \zeta^*(\tilde{\theta}) - \delta]$, we know it contains a convergent subsequence $\{M_\omega\}$ such that $\zeta^{N_\delta}(\tilde{\theta}) \to \zeta^*(\tilde{\theta}) - \omega$, where $\omega \geq \delta$ is some constant.

For any M_ω, define $k_j^{M_\omega}$ as the size of group j of the robust equilibrium of the game $\Gamma^R(M_\omega, w_2, \alpha_1, \alpha_2, F(\cdot))$, and define $x_j^{M_\omega}$ as the threshold of group j in the robust equilibrium. Denote the robust equilibrium by Π^{M_ω}. Since $\zeta^{M_\omega}(\tilde{\theta}) \to \zeta^*(\tilde{\theta}) - \omega$, it must be the case that: $\frac{\sum_{j=2}^{\bar{J}^{M_\omega}} k_j \mathbb{I}\{x_j \geq \tilde{\theta}\}}{M_\omega} \to \omega$. For ease of reference, we define:

$$k^{M_\omega} = \frac{\sum_{j=2}^{\bar{J}^{M_\omega}} k_j \mathbb{I}\{x_j \geq \tilde{\theta}\}}{M_\omega}$$

Note that a worker's payoff from applying to firm 1 (firm 2) is weakly decreasing (increasing) in people's willingness to apply to firm 1 relative to firm 2 - i.e. the payoff from applying to firm 1 (firm 2) is weakly increasing (decreasing) in $x_j^{M_\omega}$ and $k_j^{M_\omega}$. Now consider a worker i in group 1 with skill level $\theta = 0$. His payoff from applying to firm 1 in the robust equilibrium is given by:

$$u_i^{M_\omega}(1|\theta_i = 0, \Pi_{-i}^{M_\omega}) \geq (1 - \frac{1 - F(x_1^{M_\omega})}{\alpha_1 M_\omega})^{k_1^{M_\omega} - 1} (1 - \frac{1 - F(\tilde{\theta})}{\alpha_1 M_\omega})^{k^{M_\omega}} (1 - \frac{1}{\alpha_1 M_\omega})^{M_\omega - k_1^{M_\omega} - k^{M_\omega}}$$

$$u_i^{M_\omega}(2|\theta_i = 0, \Pi_{-i}^{M_\omega}) \leq w_2 (1 - \frac{F(x_1^{M_\omega})}{\alpha_2 M_\omega})^{k_1^{M_\omega} - 1} (1 - \frac{F(\tilde{\theta})}{\alpha_2 M_\omega})^{k^{M_\omega}}$$

By definition of the equilibrium:

$$(1 - \frac{1 - F(x_1^{M_\omega})}{\alpha_1 M_\omega})^{M_\omega - 1} = w_2$$

Combined with the fact that $\frac{k_1^N}{N} \to \frac{\alpha_2}{\alpha_1+\alpha_2}$:

$$
\begin{aligned}
(1-\frac{1-F(x_1^{M_\omega})}{\alpha_1 M_\omega})^{k_1^{M_\omega}-1} &\to w_2^{\frac{\alpha_2}{\alpha_2+\alpha_1}} \\
w_2(1-\frac{F(x_1^{M_\omega})}{\alpha_2 M_\omega})^{k_1^{M_\omega}-1} &\to w_2^{\frac{\alpha_2}{\alpha_2+\alpha_1}} e^{-\frac{1}{\alpha_1+\alpha_2}} \\
(1-\frac{F(\tilde{\theta})}{\alpha_2 M_\omega})^{k^{M_\omega}} &\to e^{\frac{-\omega F(\tilde{\theta})}{\alpha_2}} \\
(1-\frac{1-F(\tilde{\theta})}{\alpha_1 M_\omega})^{k^{M_\omega}} &\to e^{-\omega\frac{1-F(\tilde{\theta})}{\alpha_1}} \\
(1-\frac{1}{\alpha_1 M_\omega})^{M_\omega-k_1^{M_\omega}-k^{M_\omega}} &\to e^{\frac{\delta}{\alpha_1}-\frac{1}{\alpha_1+\alpha_2}}
\end{aligned}
$$

Therefore, as M_ω goes to infinity:

$$
\begin{aligned}
u_i^{M_\omega}(1|\theta_i=0,\Pi_{-i}^{M_\omega}) &\geq (1-\frac{1-F(x_1^{M_\omega})}{\alpha_1 M_\omega})^{k_1^{M_\omega}-1}(1-\frac{1-F(\tilde{\theta})}{\alpha_1 M_\omega})^{k^{M_\omega}}(1-\frac{1}{\alpha_1 M_\omega})^{M_\omega-k_1^{M_\omega}-k^{M_\omega}} \\
&\to w_2^{\frac{\alpha_2}{\alpha_1+\alpha_2}} e^{\frac{F(\tilde{\theta})\omega}{\alpha_1}-\frac{1}{\alpha_1+\alpha_2}} \\
&> w_2^{\frac{\alpha_2}{\alpha_1+\alpha_2}} e^{-\frac{1}{\alpha_1+\alpha_2}} \\
u_i^{M_\omega}(2|\theta_i=0,\Pi_{-i}^{M_\omega}) &\leq w_2(1-\frac{F(x_1^{M_\omega})}{\alpha_2 M_\omega})^{k_1^{M_\omega}-1}(1-\frac{F(\tilde{\theta})}{\alpha_2 M_\omega})^{k^{M_\omega}} \\
&\to w_2^{\frac{\alpha_2}{\alpha_1+\alpha_2}} e^{-\frac{F(\tilde{\theta})\omega}{\alpha_2}-\frac{1}{\alpha_1+\alpha_2}} \\
&< w_2^{\frac{\alpha_2}{\alpha_1+\alpha_2}} e^{-\frac{1}{\alpha_1+\alpha_2}}
\end{aligned}
$$

Therefore, for M_ω, $u_i^{M_\omega}(1|\theta_i=0,\Pi_{-i}^{M_\omega}) > u_i^{M_\omega}(2|\theta_i=0,\Pi_{-i}^{M_\omega})$. Meaning that worker i would want to deviate to applying for firm 1 when his skill is 0 (or extremely low since payoffs are continuous in θ_i). However, because Π^{M_ω} features only monotonic strategies by construction, this creates a contradiction.

As a result, $\forall\theta\in[a,b], \zeta^N(\theta)\to\zeta^*(\theta)$. Note that this also implies that for any $\tilde{\theta}\in(a,x_1)$, $\frac{\sum_{j=2}^{\bar{J}_N} k_{j,N}\mathbb{I}\{x_{j,N}\geq\tilde{\theta}\}}{N}\to 0$.

□

Proposition 9:

Proof. Consider worker i with skill realisation θ_i and a robust equilibrium of the replicator economy $\Pi^{R,N}$. Assume W.L.O.G. that $\theta_i\in[x_{j,N},x_{j-1,N}]$ for some $j\in\{1,...,\bar{J}_N\}$. We define $x_0=b$.

Let $k_{j,N}$ be the size of group j in the $\Pi^{R,N}$ equilibrium. Agent i faces the following payoff:

$$u_i(1|\theta_i,\Pi^{R,N}_{-i}) = \frac{1}{1-\frac{1-F(\max\{\theta_i,x_{JR(i),N}\})}{H_1}}\left[\prod_{h=1}^{j-1}(1-\frac{1-F(x_{h,N})}{H_1})^{k_{h,N}}\prod_{h=j}^{\bar{J}_N}(1-\frac{1-F(\theta_i)}{H_1})^{k_{h,N}}\right]$$

$$u_i(2|\theta_i,\Pi^{R}_{-i}) = w_2\frac{1}{1-\frac{F(x_{JR(i),N})-F(\min\{\theta_i,x_{JR(i),N}\})}{H_2}}\left[\prod_{h=1}^{j-1}(1-\frac{F(x_{h,N})-F(\theta_i)}{H_2})^{k_{h,N}}\right]$$

First consider $\theta_i > x_1 = F^{-1}(1+\alpha_1\ln(w_2))$. Then for all N sufficiently large, $\theta_i \in [x_{1,N},b]$, and $\pi^{R,N}_j(\theta_j) = 1\forall j$, and $\forall\theta_j \geq \theta_i$. Worker i's payoff is therefore given by:

$$\begin{aligned}
u^*_i(\theta_i|\Pi^{R,N}) =& u_i(1|\theta_i,\Pi^{R,N}_{-i}) \\
&= (1-\frac{1-F(\theta_i)}{H_1})^{N-1} \\
&= \left[(1-\frac{1-F(\theta_i)}{\alpha_1 N})^{\frac{\alpha_1 N}{1-F(\theta_i)}}\right]^{\frac{1-F(\theta_i)}{\alpha_1}} \\
&\to e^{-\frac{1-F(\theta_i)}{\alpha_1}} \quad \forall\theta_i > x_1
\end{aligned}$$

Note that by the proof of Proposition 8, for any $\theta_i \in (a,x_1)$:

$$\lim_{N\to\infty}\frac{\sum_{h=2}^{\bar{J}_N}k_{h,N}\mathbb{I}\{x_{h,N}\geq\theta_i\}}{N} \to 0$$

Now consider $\theta_i < x_1$. For all N sufficiently large, it must be the case that $\theta_i \in [x_{j,N},x_{j-1,N}]$ for some $j \in \{2,...,\bar{J}_N\}$. Therefore, if worker i applies to w_1, then his payoff is:

$$\begin{aligned}
u_i(1|\theta_i,\Pi^{R,N}_{-i}) &= \frac{1}{1-\frac{1-F(\max\{\theta_i,x_{JR(i),N}\})}{H_1}}\left[\prod_{h=1}^{j-1}(1-\frac{1-F(x_{h,N})}{H_1})^{k_{h,N}}\prod_{h=j}^{\bar{J}_N}(1-\frac{1-F(\theta_i)}{H_1})^{k_{h,N}}\right] \\
&\to \lim_{N\to\infty}\left[(1-\frac{1-F(x_{1,N})}{\alpha_1 N})^{\frac{\alpha_1 N}{1-F(x_{1,N})}}\right]^{\frac{k_{1,N}(1-F(x_{1,N}))}{\alpha_1 N}}\lim_{N\to\infty}\left[(1-\frac{1-F(\theta_i)}{\alpha_1 N})^{\frac{\alpha_1 N}{1-F(x_{\theta_i})}}\right]^{\frac{(1-F(\theta_i))(N-\sum_{h=1}^{j-1}k_{h,N})}{\alpha_1 N}} \\
&= \left(\frac{1}{e}\right)^{\lim_{N\to\infty}\frac{k_{1,N}(1-F(x_{1,N}))}{\alpha_1 N}}\left(\frac{1}{e}\right)^{\lim_{N\to\infty}\frac{1-F(\theta_i)(N-\sum_{h=1}^{j-1}k_{h,N})}{\alpha_1 N}} \\
&= \left(\frac{1}{e}\right)^{\frac{1-F(x_1)\alpha_2}{\alpha_1(\alpha_1+\alpha_2)}}\left(\frac{1}{e}\right)^{\frac{(1-F(\theta_i))}{\alpha_1+\alpha_2}} \\
&= \left(\frac{1}{e}\right)^{\frac{-\alpha_1\ln(w_2)\alpha_2}{\alpha_1(\alpha_1+\alpha_2)}}\left(\frac{1}{e}\right)^{\frac{1-F(\theta_i)}{\alpha_1+\alpha_2}} \\
&= w_2^{\frac{\alpha_2}{\alpha_1+\alpha_2}}e^{-\frac{1-F(\theta_i)}{\alpha_1+\alpha_2}}
\end{aligned}$$

Similarly, if worker i applies to w_2, then his payoff is:

$$
\begin{aligned}
u_i(2|\theta_i,\Pi^R_{-i}) &= w_2 \frac{1}{1-\frac{F(x_{JR(i),N})-F(\min\{\theta_i,x_{JR(i),N}\})}{H_2}} \left[\prod_{h=1}^{j-1}(1-\frac{F(x_{h,N})-F(\theta_i)}{H_2})^{k_{h,N}}\right] \\
&\to w_2 \lim_{N\to\infty}\left[(1-\frac{F(x_{1,N})-F(\theta_i)}{\alpha_2 N})^{\frac{\alpha_2 N}{F(x_{1,N})-F(\theta_i)}}\right]^{k_{1,N}\frac{F(x_{1,N})-F(\theta_i)}{\alpha_2 N}} \\
&= w_2\left(\frac{1}{e}\right)^{\lim_{N\to\infty} k_{1,N}\frac{F(x_{1,N})-F(\theta_i)}{\alpha_2 N}} \\
&= w_2\left(\frac{1}{e}\right)^{\frac{\alpha_2(F(x_1)-F(\theta_i))}{\alpha_2(\alpha_1+\alpha_2)}} \\
&= w_2\left(\frac{1}{e}\right)^{\frac{1+\alpha_1\ln(w_2)-F(\theta_i)}{\alpha_1+\alpha_2}} \\
&= w_2^{\frac{\alpha_1}{\alpha_1+\alpha_2}}\left(\frac{1}{e}\right)^{\frac{1-F(\theta_i)}{\alpha_1+\alpha_2}}
\end{aligned}
$$

□

Therefore $\forall i$:

$$
u_i^*(\theta_i|\Pi^{R,N}) \to \begin{cases} e^{-\frac{1-F(\theta_i)}{\alpha_1}} & \text{if } \theta_i \geq x_1 = F^{-1}(1+\alpha_1\ln(w_2)) \\ w_2^{\frac{\alpha_1}{\alpha_1+\alpha_2}}\left(\frac{1}{e}\right)^{\frac{1-F(\theta_i)}{\alpha_1+\alpha_2}} & \text{if else} \end{cases}
$$

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